

Generative Continuation Geometry

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Chapter 1

Continuation Attractors

A fern frond is not drawn leaf by leaf. Look closely at one pinna and it is, near enough, a smaller fern; look closer still at one pinnule of that pinna and it is, near enough, a smaller fern again. Nobody specified the shape of the whole plant directly. A small number of growth rules, applied to their own output over and over, produced something with more detail than any of the rules themselves contain. The shape is not stored anywhere. It is generated.

Continuation Geometry spent its length measuring futures: how much room a state has, how far apart two states' futures sit, when a distinction between futures collapses to nothing. All of that took the space of futures as given, something to be measured once it already existed. This book asks a different question. Can a space of futures itself be grown, the way the fern is grown, from a small number of simple moves applied to their own output? If continuation spaces can be generated rather than merely measured, the natural unit of study stops being the space and becomes the small set of operations that built it.

This opening chapter answers that question in the one case where the answer is already, in a precise sense, classical. It is not a new theorem. It is an old and beautiful one, due to Hutchinson, applied here to the specific metric geometry the first book already built. The payoff is that self-similar continuation structures, the kind suggested by a duck, a crown, and a leaf all arising from the same underlying rule, turn out to rest on exactly the Hausdorff-distance machinery already in hand, with nothing new invented and nothing smuggled in from outside.

Preliminaries: expansion is not generation

Two facts from *Continuation Geometry* are worth restating here, briefly and as what they actually are: lemmas, not theorems, and useful precisely because they show what generation is *not*.

Lemma 1.1 (Reachable expansion). *For $S_t \subseteq X$ and $\text{Adj}(S_t) := \bigcup_{x \in S_t} \Phi(x)$ under a one-step reachability map Φ , the sequence $S_{t+1} := S_t \cup \text{Adj}(S_t)$ satisfies $S_t \subseteq S_{t+1}$ for every t .*

Proof. Immediate: $S_t \subseteq S_t \cup \text{Adj}(S_t) = S_{t+1}$ by definition of union. □

Remark 1.1. *This, together with the reachable-set recursion proved by induction in *Continuation Geometry* (Kauffman's adjacent possible as an instance of finite-horizon reachability), is the full content of what that book called generation, and it is worth being honest that it earns the name only loosely. The*

sequence $S_0, S_1, S_2, \dots$ grows monotonically and without bound unless the underlying space is finite; nothing about it converges, and nothing about it is self-similar. It is accumulation, not generation in the sense the fern illustrates. What follows is different in kind: rather than a sequence of ever-larger unions, it is a fixed operator applied repeatedly to itself, and the interesting behavior is not growth but convergence to a shape that reproduces its own generating rule.

The space of shapes

Fix $X = \mathbb{R}^n$ (or any complete metric space) and let $\mathcal{K}(X)$ denote the set of nonempty compact subsets of X , equipped with the Hausdorff distance d_H of Chapter 6 of *Continuation Geometry*.

Theorem 1.1 (Completeness of the space of shapes). *If (X, d) is complete, then $(\mathcal{K}(X), d_H)$ is complete.*

Proof sketch. Given a Cauchy sequence $\{A_n\}$ in $(\mathcal{K}(X), d_H)$, define $A := \bigcap_{N \geq 1} \overline{\bigcup_{n \geq N} A_n}$. One shows, using completeness of X and the Cauchy property of $\{A_n\}$ in d_H , that A is nonempty and compact, and that $d_H(A_n, A) \rightarrow 0$. This is a standard but genuinely technical result in its full form, requiring care with total boundedness and a diagonal argument to extract convergent selections from each A_n ; the complete argument is given in [2] and [3] and is not reproduced in full here. $\square$

Remark 1.2. *This theorem is doing real work that is easy to take for granted. Without it, nothing built on top of $(\mathcal{K}(X), d_H)$ would be guaranteed to have a limit at all; a sequence of shapes could get closer and closer together, in the sense of Hausdorff distance, without converging to any actual shape. That this cannot happen, that the space of compact reachable sets is itself complete whenever the underlying state space is, is the entire reason the construction below is guaranteed to produce something rather than merely getting closer to nothing in particular.*

Contractions and the Hutchinson operator

Definition 1.1 (Contraction). *$T : X \rightarrow X$ is a contraction with constant $c \in [0, 1)$ if $d(Tx, Ty) \leq c d(x, y)$ for all $x, y \in X$.*

Lemma 1.2 (Contractions act contractively on shapes). *If T is a contraction on X with constant c , then for $A, B \in \mathcal{K}(X)$, $d_H(T(A), T(B)) \leq c d_H(A, B)$, where $T(A) := \{Ta : a \in A\}$.*

Proof. Fix $a \in A$. Since B is compact, there is $b \in B$ with $d(a, b) = \text{dist}(a, B) \leq d_H(A, B)$. Then $d(Ta, Tb) \leq c d(a, b) \leq c d_H(A, B)$, and since $Tb \in T(B)$, $\text{dist}(Ta, T(B)) \leq c d_H(A, B)$. Taking the supremum over $a \in A$ gives $\sup_{a' \in T(A)} \text{dist}(a', T(B)) \leq c d_H(A, B)$. The symmetric argument, exchanging the roles of A and B , gives the same bound the other way, so $d_H(T(A), T(B)) \leq c d_H(A, B)$. $\square$

Lemma 1.3 (Unions are 1-Lipschitz). *For finite collections $\{A_i\}_{i=1}^n, \{B_i\}_{i=1}^n$ of nonempty compact sets, $d_H(\bigcup_i A_i, \bigcup_i B_i) \leq \max_i d_H(A_i, B_i)$.*

Proof. Let $m := \max_i d_H(A_i, B_i)$. For $x \in \bigcup_i A_i$, $x \in A_j$ for some j , so $\text{dist}(x, B_j) \leq d_H(A_j, B_j) \leq m$, and since $B_j \subseteq \bigcup_i B_i$, $\text{dist}(x, \bigcup_i B_i) \leq \text{dist}(x, B_j) \leq m$. Taking the supremum over $x \in \bigcup_i A_i$ gives $\sup_x \text{dist}(x, \bigcup_i B_i) \leq m$. The symmetric argument, exchanging the roles of the two collections, gives the same bound the other way, so $d_H(\bigcup_i A_i, \bigcup_i B_i) \leq m$. $\square$

Definition 1.2 (Hutchinson operator). Given contractions $T_1, \dots, T_n$ on X with constants $c_1, \dots, c_n$, define $\mathcal{H} : \mathcal{K}(X) \rightarrow \mathcal{K}(X)$ by $\mathcal{H}(A) := \bigcup_{i=1}^n T_i(A)$.

Proposition 1.1 ($\mathcal{H}$ is a contraction on shapes). $d_H(\mathcal{H}(A), \mathcal{H}(B)) \leq c d_H(A, B)$ for $c := \max_i c_i < 1$.

Proof. By Lemma 1.3, $d_H(\mathcal{H}(A), \mathcal{H}(B)) = d_H(\bigcup_i T_i(A), \bigcup_i T_i(B)) \leq \max_i d_H(T_i(A), T_i(B))$. By Lemma 1.2, $d_H(T_i(A), T_i(B)) \leq c_i d_H(A, B)$ for each i . Hence $d_H(\mathcal{H}(A), \mathcal{H}(B)) \leq \max_i c_i d_H(A, B) = c d_H(A, B)$. $\square$

The Continuation Attractor Theorem

Theorem 1.2 (Continuation Attractor Theorem, after Hutchinson). Let $T_1, \dots, T_n$ be contractions on a complete metric space X with constants $c_1, \dots, c_n$, and let $c := \max_i c_i < 1$. Then there exists a unique $A^* \in \mathcal{K}(X)$ with $\mathcal{H}(A^*) = A^*$, and for any $A_0 \in \mathcal{K}(X)$,

$$d_H(\mathcal{H}^k(A_0), A^*) \leq \frac{c^k}{1-c} d_H(A_0, \mathcal{H}(A_0)) \xrightarrow{k \rightarrow \infty} 0.$$

Proof. By Theorem 1.1, $(\mathcal{K}(X), d_H)$ is complete. By Proposition 1.1, $\mathcal{H}$ is a contraction on this complete metric space. The Banach Fixed Point Theorem [1] then supplies a unique fixed point A^* and the stated geometric convergence rate, applied directly to $\mathcal{H}$ on $(\mathcal{K}(X), d_H)$. $\square$

Remark 1.3. No part of this proof required inventing anything: completeness of $\mathcal{K}(X)$ is Theorem 1.1, the contraction property of $\mathcal{H}$ is Proposition 1.1 built from two short lemmas, and the conclusion is the Banach Fixed Point Theorem, unmodified. This is the sense in which this chapter is mathematically conservative even though its subject, self-similar generated shape, sounds exotic: every piece was already sitting in the metric structure Chapter 6 of Continuation Geometry built for an entirely different purpose.

A worked attractor

Example 1.1 (Two refusals). Let $X = [0, 1]$, and let $T_1(x) := x/3$, $T_2(x) := x/3 + 2/3$, both contractions with constant $c = 1/3$. The attractor A^* satisfying $A^* = T_1(A^*) \cup T_2(A^*)$ is the standard middle-thirds Cantor set.

Remark 1.4. Read in the vocabulary of continuation, T_1 and T_2 are the two survivors of a repeated refusal: at every stage, whatever occupies the middle third of the current set of admissible outcomes is discarded, and the remaining two-thirds are each rescaled to stand in for the whole. What survives in the limit is neither of the two pieces nor their simple union at any finite stage, but the unique shape that reproduces itself exactly under the same rule that built it, uncountable, nowhere dense, and entirely determined by two contraction maps. Whether operators named Collapse or Refuse elsewhere in this corpus are literally instances of T_i in this technical sense, maps satisfying the contraction inequality on some genuine underlying metric, is a claim about those specific constructions and is not asserted here; what is established is only that if a family of continuation-narrowing operators can be shown to be contractions on $(\mathcal{K}(X), d_H)$, this theorem applies to them without modification.

What this chapter does not yet claim

This chapter establishes that self-similar continuation structures are mathematically available, not that any particular structure this corpus has described informally is an instance of one. Two gaps in particular are worth naming rather than leaving implicit. First, nothing here says which operators arising from an actual continuation problem, as opposed to a hand-picked pair of affine maps on $[0, 1]$, are genuinely contractions on a genuine metric; that has to be checked case by case, the way it would for any application of a fixed-point theorem. Second, and more speculatively, the question of whether a small, fixed family of primitive operators could generate *every* admissible continuation geometry to arbitrary Hausdorff-distance accuracy, a continuation-geometric analogue of universal approximation, is not addressed by this theorem and is not claimed here. It is a genuine open question, and answering it is a natural place for this book to go next, once the machinery in the chapters between here and there, quotient maps and operator composition among them, has been built on equally solid ground.

Chapter 2

Continuation Quotients

Two rivers can rise on opposite sides of a mountain range, cross entirely different terrain, pass through valleys that share nothing in common, and still arrive at the same confluence and become, from that point on, one river. Everything about their separate histories, which rocks each one wore smooth, which tributaries fed it, how long each took to descend, becomes irrelevant downstream of the merge. A boat launched below the confluence cannot tell which branch its water came from. Not because the information was hidden. Because it no longer makes a difference to anything the boat will experience from here on.

The previous chapter showed that a continuation space can be grown from a small number of generating operators. The natural next question is what happens when two different generating histories, different sequences of operators, different starting points, produce the same continuation space at the end. When does that happen, and what, if anything, does it mean? The temptation is to reach immediately for language like information loss or memory collapse, language already used loosely elsewhere in this corpus. This chapter resists that temptation on purpose. It defines the relevant object first, proves what can be proved about it without reference to any interpretation, and only afterward asks what, if anything, has been lost.

The continuation quotient

Definition 2.1 (Continuation quotient). *Let H be a set of histories and let $\Gamma : H \rightarrow \mathcal{K}(X)$ assign to each history its continuation space (or, following Chapter 6 of Continuation Geometry, its finite-horizon reachable set R_T for a fixed horizon T). Define $h_1 \sim_\Gamma h_2$ iff $\Gamma(h_1) = \Gamma(h_2)$. This is an equivalence relation (reflexivity, symmetry, and transitivity all following immediately from equality of sets), and the continuation quotient is $H/\sim_\Gamma$, with quotient map $q : H \rightarrow H/\sim_\Gamma$ sending h to its equivalence class $[h]$.*

Nothing has been said yet about distinctions, memory, or loss. This is deliberate: the object exists, and has structure, independently of any of those interpretations.

Proposition 2.1 (Well-defined continuation projection). *There exists a unique map $\bar{\Gamma} : H/\sim_\Gamma \rightarrow \mathcal{K}(X)$ such that $\Gamma = \bar{\Gamma} \circ q$.*

Proof. Define $\bar{\Gamma}([h]) := \Gamma(h)$. This is well-defined: if $[h_1] = [h_2]$, then $h_1 \sim_\Gamma h_2$, so $\Gamma(h_1) = \Gamma(h_2)$ by definition of $\sim_\Gamma$, so the value assigned to $[h_1] = [h_2]$ does not depend on which representative was chosen. By construction, $\bar{\Gamma}(q(h)) = \bar{\Gamma}([h]) = \Gamma(h)$ for every $h \in H$, so

$\Gamma = \bar{\Gamma} \circ q$. For uniqueness, suppose $\bar{\Gamma}'$ also satisfies $\Gamma = \bar{\Gamma}' \circ q$. Since q is surjective, every element of $H/\sim_\Gamma$ is $q(h)$ for some h , and $\bar{\Gamma}'(q(h)) = \Gamma(h) = \bar{\Gamma}(q(h))$ for every such h , so $\bar{\Gamma}' = \bar{\Gamma}$. $\square$

Remark 2.1. *This is routine, and it is worth saying plainly that it is routine: it is the universal property of a quotient set, applied to nothing more exotic than the map Γ itself. What makes it worth stating is not its difficulty but what it licenses: from here on, $\bar{\Gamma}$ is a genuine function on the quotient, not merely a suggestive notation, and anything proved about $\bar{\Gamma}$ is a fact about the quotient set itself, not about any particular history used to reach a point in it.*

A structural lemma

The worked example below relies on the following fact about time-homogeneous, deterministic-image dynamics, worth isolating here.

Lemma 2.1 (One-step agreement propagates). *Suppose $R_1(x) = R_1(y)$ for states x, y under a time-homogeneous reachability map. Then $R_T(x) = R_T(y)$ for every $T \geq 1$.*

Proof. By induction on T . The case $T = 1$ is the hypothesis. Suppose $R_T(x) = R_T(y)$. Since the dynamics are time-homogeneous, $R_{T+1}(x) = \bigcup_{z \in R_T(x)} R_1(z)$ and likewise for y ; since $R_T(x) = R_T(y)$, these unions range over the same set of states z , so $R_{T+1}(x) = R_{T+1}(y)$. $\square$

Remark 2.2. *This is the precise sense in which a collision, once it happens, is permanent under time-homogeneous dynamics: two states whose immediate futures coincide exactly will have every subsequent future coincide as well, since there is nothing left in either state's future to distinguish it from the other's.*

Quotient invariants

Proposition 2.2 (Quotient invariance). *Let $F : \mathcal{K}(X) \rightarrow Y$ be any function. Then $F \circ \Gamma : H \rightarrow Y$ factors through q : there is a unique $\bar{F} : H/\sim_\Gamma \rightarrow Y$ with $F \circ \Gamma = \bar{F} \circ q$.*

Proof. $F \circ \Gamma : H \rightarrow Y$ satisfies $(F \circ \Gamma)(h_1) = (F \circ \Gamma)(h_2)$ whenever $h_1 \sim_\Gamma h_2$, since $\Gamma(h_1) = \Gamma(h_2)$ by definition of $\sim_\Gamma$ and F is a function, hence single-valued. Apply Proposition 2.1 with $F \circ \Gamma$ in place of Γ to obtain the factorization. $\square$

Corollary 2.1. *Continuation volume Vol_T , the distinction margin $A_{\mathcal{D}}^T$ relative to any fixed reference distinction, and boundary proximity δ , all defined in Continuation Geometry as functions of a reachable set or of a state's position relative to one, descend to well-defined functions on $H/\sim_\Gamma$.*

Proof. Each is of the form $F(\Gamma(h))$ or $F(R_T(x(h)))$ for a function F depending only on the reachable set itself (Lebesgue measure of the set, Hausdorff distance to another fixed set, ambient distance from a fixed reference set), so Proposition 2.2 applies directly to each. $\square$

Remark 2.3. *This is a broader statement than it might first appear. It says that any quantity this book or its predecessor has defined, or will define, purely in terms of a reachable set, without reference to which history produced that set, is automatically constant on $\sim_\Gamma$ -equivalence classes, with no further argument required each time a new such quantity is introduced. The work of checking invariance, done once here, does not need to be redone for every future functional built the same way.*

Finite checkability

Lemma 2.1 shows that equal one-step images propagate forward forever. It says nothing about the converse: two states whose one-step images differ can still coincide at some later horizon. Nothing in this chapter has claimed otherwise, and the general question, whether continuation equivalence at every horizon can still be checked in finitely many steps despite this, deserves a correct answer rather than an implied one.

Proposition 2.3 (Eventual periodicity bounds the check). *If X is finite, then for any $x, y \in X$ the paired sequence $(R_1(x), R_1(y)), (R_2(x), R_2(y)), \dots$ is eventually periodic, with both the pre-period and the period bounded by $4^{|X|}$. Consequently, whether $R_T(x) = R_T(y)$ holds for all sufficiently large T can be determined by comparing $R_T(x)$ and $R_T(y)$ for T ranging only up to $2 \cdot 4^{|X|}$.*

Proof. Each pair $(R_T(x), R_T(y))$ is an element of $2^X \times 2^X$, which has $4^{|X|}$ elements, so the paired sequence takes at most $4^{|X|}$ distinct values; among its first $4^{|X|} + 1$ terms, two must coincide by pigeonhole, $(R_p(x), R_p(y)) = (R_q(x), R_q(y))$ for some $1 \leq p < q \leq 4^{|X|} + 1$. Time-homogeneity means $R_{T+1} = \Psi(R_T)$ for the fixed map $\Psi(S) := \bigcup_{z \in S} \Phi(z)$, applied identically to the x and y components, so equality of the pair at p and q propagates: $(R_{p+j}(x), R_{p+j}(y)) = (\Psi^j(R_p(x)), \Psi^j(R_p(y))) = (\Psi^j(R_q(x)), \Psi^j(R_q(y))) = (R_{q+j}(x), R_{q+j}(y))$ for every $j \geq 0$: the paired sequence is periodic with period dividing $q - p \leq 4^{|X|}$ from index $p \leq 4^{|X|}$ onward. Comparing the pair over one further full period, at most $4^{|X|}$ more steps beyond the pre-period bound, settles whether x and y agree from that point on. $\square$

Remark 2.4. *Bounding each sequence's own period separately and then comparing them over a single shared window is not sufficient on its own: two sequences with different individual periods p_x, p_y need not realign within $\max(p_x, p_y)$ steps, only within $\text{lcm}(p_x, p_y)$ steps, which can exceed either period substantially. Working with the pair from the start avoids this, since the pair's own period already accounts for however the two component periods interact.*

Remark 2.5. *This is the correct version of the claim that continuation equivalence is finitely checkable. It is a coarser bound than a sharp analysis of any particular system would give, since $2^{|X|}$ is a worst case over all possible dynamics on a state space of that size, but it is unconditional: no structural assumption on Φ beyond finiteness of X and time-homogeneity is needed. The traffic light below reaches its own eventual behavior in four steps, far inside this bound, and it is worth seeing exactly how.*

The traffic light, revisited

Recall the traffic light of *Continuation Geometry*, Chapter 10: states $(k, r) \in \{0, 1, 2, 3\} \times \{0, 1\}$, phase k and pending request r . To compute Γ properly requires treating the system's branching explicitly rather than leaving it implicit. Let $e \in \{0, 1\}$ record whether a new pedestrian request arrives during a given step (exogenous, not controlled by the phase). The transition is

$$\Phi((k, r), e) = ((k + 1) \bmod 4, r'), \quad r' = \begin{cases} 0 & k + 1 \equiv 0 \pmod{4} \text{ (clear on arrival at red)} \\ 1 & k + 1 \not\equiv 0 \pmod{4} \text{ and } (r = 1 \text{ or } e = 1) \\ 0 & k + 1 \not\equiv 0 \pmod{4} \text{ and } r = 0 \text{ and } e = 0. \end{cases}$$

Let H be the set of finite paths of the system, and $\Gamma(h) := R_1(x(h))$, the one-step reachable set from the endpoint of h , ranging over $e \in \{0, 1\}$.

Proposition 2.4 (Exact computation of the traffic-light quotient at $T = 1$). *Under $\sim_\Gamma$, the eight states fall into exactly seven classes: the single nontrivial class $\{(3, 0), (3, 1)\}$, and six singleton classes, one for each remaining state.*

Proof. Compute R_1 for all eight states directly from Φ . For $k \neq 3$, arrival is at a nonzero phase and the ordinary branching rule applies, $r' = 1$ iff $r = 1$ or $e = 1$:

$$\begin{aligned} R_1(0, 0) &= \{(1, 0), (1, 1)\}, & R_1(0, 1) &= \{(1, 1)\}, \\ R_1(1, 0) &= \{(2, 0), (2, 1)\}, & R_1(1, 1) &= \{(2, 1)\}, \\ R_1(2, 0) &= \{(3, 0), (3, 1)\}, & R_1(2, 1) &= \{(3, 1)\}. \end{aligned}$$

For $k = 3$, arrival is at phase 0, so clearing fires unconditionally regardless of r or e :

$$R_1(3, 0) = \{(0, 0)\}, \quad R_1(3, 1) = \{(0, 0)\}.$$

Comparing all eight sets directly: the three size-two sets, $\{(1, 0), (1, 1)\}$, $\{(2, 0), (2, 1)\}$, $\{(3, 0), (3, 1)\}$, are pairwise distinct; the four remaining size-one sets, $\{(1, 1)\}$, $\{(2, 1)\}$, $\{(0, 0)\}$ (appearing twice, from $(3, 0)$ and $(3, 1)$), and $\{(0, 0)\}$ again, give exactly one coincidence, $R_1(3, 0) = R_1(3, 1) = \{(0, 0)\}$, and no other. $\square$

Now define the mechanical quotient $\sim_M$ independently of Γ , as the identification the design actually performs: the clearing rule fires on the transition *into* phase 0, so it is the two states about to make that transition, $(3, 0)$ and $(3, 1)$, that the design treats as interchangeable, both being followed unconditionally by $(0, 0)$.

Proposition 2.5 (The design is exact at $T = 1$). $\sim_M = \sim_\Gamma$.

Proof. Both relations have exactly one nontrivial class, $\{(3, 0), (3, 1)\}$, and agree as singletons on every other state: $\sim_M$ by definition of the clearing rule, $\sim_\Gamma$ by Proposition 2.4. Two equivalence relations on the same finite set with identical partitions are equal. $\square$

Remark 2.6. *It is worth being precise about which of two similar-sounding designs this result actually vindicates. Clearing on arrival at phase 0 (the rule above) is exact. A superficially similar alternative, clearing whenever the system is currently at phase 0 rather than arriving there, would instead identify $(0, 0)$ and $(0, 1)$. But $R_1(0, 0) = \{(1, 0), (1, 1)\} \neq \{(1, 1)\} = R_1(0, 1)$: a state at the red light with no pending request can still branch toward either outcome next, while one with a pending request cannot, precisely because that request is about to be served. The alternative design would merge two states with genuinely different one-step futures, exactly the coarser, information-destroying case this chapter opened by distinguishing from the exact one. The two designs differ only in whether clearing is keyed to the transition or to the state, and only one of them is correct.*

What one horizon hides

Restricting Proposition 2.4 to $T = 1$ was not a simplification; it is what Γ was defined to measure. But Proposition 2.3 guarantees the longer-horizon question has a definite, finitely checkable answer, and for this system that answer is worth computing directly rather than left as an abstract guarantee.

Proposition 2.6 (The request bit is eventually redundant everywhere). *For each phase k , the pair $(k, 0)$ and $(k, 1)$ satisfies $R_T(k, 0) = R_T(k, 1)$ for all $T \geq 4 - k$, and not before.*

Proof by direct computation. Compute $R_T(k, 0)$ and $R_T(k, 1)$ for $T = 0, \dots, 4 - k$ at each phase directly from Ψ , rather than inferring later horizons from earlier ones: a size mismatch at one horizon does not, on its own, preclude equality reappearing or vanishing at a different horizon, so each phase's first point of equality is verified independently here rather than propagated.

For $k = 3$: $R_0(3, 0) = \{(3, 0)\} \neq \{(3, 1)\} = R_0(3, 1)$; $R_1(3, 0) = R_1(3, 1) = \{(0, 0)\}$ (Proposition 2.4). First equality at $T = 1 = 4 - 3$.

For $k = 2$: $R_0(2, 0) = \{(2, 0)\} \neq \{(2, 1)\} = R_0(2, 1)$; $R_1(2, 0) = \{(3, 0), (3, 1)\} \neq \{(3, 1)\} = R_1(2, 1)$ (both sets computed directly from Φ , not equal since one has two elements and the other one); $R_2(2, 0) = \Psi(\{(3, 0), (3, 1)\}) = \{(0, 0)\} = \Psi(\{(3, 1)\}) = R_2(2, 1)$. First equality at $T = 2 = 4 - 2$.

For $k = 1$: R_0, R_1 inequality follows the same way (cardinality 2 versus 1 at $T = 1$, distinct singletons at $T = 0$); $R_2(1, 0) = \{(3, 0), (3, 1)\} \neq \{(3, 1)\} = R_2(1, 1)$, still unequal; $R_3(1, 0) = \Psi(\{(3, 0), (3, 1)\}) = \{(0, 0)\} = \Psi(\{(3, 1)\}) = R_3(1, 1)$. First equality at $T = 3 = 4 - 1$.

For $k = 0$: $R_0(0, 0) = \{(0, 0)\} \neq \{(0, 1)\} = R_0(0, 1)$, distinct singletons; R_1, R_2, R_3 each pit a two-element set against a one-element set, as computed directly from Φ at each step, so none of the three can coincide; $R_4(0, 0) = \{(0, 0)\} = R_4(0, 1)$. First equality at $T = 4 = 4 - 0$.

In every case the inequality at each horizon strictly before $4 - k$ was checked directly from the explicit sets, not inferred from the inequality at a different horizon. $\square$

Remark 2.7. *This is the sharper and more interesting fact the single-horizon result was hiding. The request bit is not redundant in general; it is redundant by a deadline, and that deadline is exactly the number of steps remaining until the light is guaranteed to cycle through its clearing transition, four steps for a state that has just left it, one step for a state about to enter it. Every state's request bit stops mattering to its continuation space at precisely the moment continuation stops being able to tell whether that request will still be pending when clearing finally arrives, and not one step before. The design clears the bit at the earliest possible moment, $T = 1$ after arrival, which is also the latest moment at which failing to clear it would still have been observably wrong; every other phase's bit becomes redundant on its own, without any design intervention, simply because the light's own periodicity eventually answers the question the bit was tracking.*

What quotients do not yet explain

This chapter has shown when two histories are interchangeable from the standpoint of every quantity built purely from continuation spaces, and it has shown, in one worked case, that a designed identification can be checked against that standard exactly rather than merely defended by intuition. It has not yet said anything about what is lost when a quotient is coarser than $\sim_\Gamma$, only that the traffic light's particular quotient happens not to be. The next chapter takes up the coarser case directly: not whether information is destroyed, which this chapter's machinery can already detect, but how much, and by what measure.

Chapter 3

Distinction Loss Under Quotients

A library catalog entry is a quotient of a book: title, author, and a handful of subject headings standing in for everything between the covers. A good catalog loses nothing a patron actually needs in order to decide whether to pull the book from the shelf. A bad one can lose quite a lot; a catalog that records only “fiction” or “nonfiction” will happily file a memoir next to a novel and a field guide next to a textbook, and a patron relying on that heading alone has no way to tell them apart until the book is already in hand. The catalog entry is not wrong, exactly; it simply was not built to preserve the distinctions this particular patron cares about.

The previous chapter showed how to check whether a given identification of histories is exact: whether it merges only histories whose continuation spaces already coincide, or whether it merges some that do not. That check was binary, exact or not. This chapter asks the sharper question a librarian actually has to answer. When an identification is not exact, how much is lost, and where exactly does the loss occur? The traffic light’s badly designed alternative from the previous chapter turns out to have a precise answer, not just a qualitative one.

Fibers and their content

Definition 3.1 (Fiber, Γ -content, loss). *For a quotient $q : H \rightarrow O$ and $o \in O$, the fiber over o is $q^{-1}(o) \subseteq H$. Its Γ -content is $C_\Gamma(o) := \{\Gamma(h) : h \in q^{-1}(o)\}$, the set of distinct continuation spaces appearing in the fiber. The loss at o is*

$$\mathcal{L}(o) := \log |C_\Gamma(o)|,$$

so $\mathcal{L}(o) = 0$ exactly when every history mapping to o under q shares a single continuation space.

Proposition 3.1 (Zero loss characterizes Γ -respecting quotients). $\mathcal{L}(o) = 0$ for every $o \in O$ if and only if q refines $\sim_\Gamma$: $h_1 \sim_q h_2 \implies h_1 \sim_\Gamma h_2$.

Proof. If q refines $\sim_\Gamma$, then $h_1 \sim_q h_2$ implies $\Gamma(h_1) = \Gamma(h_2)$, so every fiber of q lies inside a single $\sim_\Gamma$ -class and has $|C_\Gamma(o)| = 1$, giving $\mathcal{L}(o) = 0$. Conversely, if $\mathcal{L}(o) = 0$ for every o , then every fiber has $|C_\Gamma(o)| = 1$, so any two histories in the same fiber share the same Γ -value: $h_1 \sim_q h_2 \implies \Gamma(h_1) = \Gamma(h_2) \implies h_1 \sim_\Gamma h_2$, so q refines $\sim_\Gamma$. $\square$

The coarsest lossless quotient

Theorem 3.1 ($\sim_\Gamma$ is the coarsest zero-loss quotient). *Every zero-loss quotient $q : H \rightarrow O$ refines $\sim_\Gamma$, and $\sim_\Gamma$ itself has zero loss everywhere. Consequently $\sim_\Gamma$ is, up to relabeling of O , the unique coarsest quotient of H with zero loss.*

Proof. The first claim is Proposition 3.1. That $\sim_\Gamma$ itself has zero loss is immediate: by definition, $h_1 \sim_\Gamma h_2$ already means $\Gamma(h_1) = \Gamma(h_2)$, so every $\sim_\Gamma$ -class has exactly one Γ -value. Since every zero-loss q refines $\sim_\Gamma$, no zero-loss quotient can be strictly coarser than $\sim_\Gamma$, and $\sim_\Gamma$ achieves zero loss, so it is the coarsest one. $\square$

Remark 3.1. *This is the quotient-theoretic form of a fact Chapter 5 of Continuation Geometry established for sufficient statistics without this vocabulary. A sufficient statistic φ there was required to satisfy $\varphi(H_t) = \varphi(H'_t) \implies \Gamma(x_t | H_t) = \Gamma(x_t | H'_t)$, which is exactly the condition that the quotient induced by φ refines $\sim_\Gamma$. Theorem 3.1 says $\sim_\Gamma$ itself is the coarsest possible sufficient statistic: the most compressed summary of a history that still loses nothing continuation-relevant. Any coarser summary is not merely less detailed; by Proposition 3.1, it has strictly positive loss somewhere.*

Proposition 3.2 (Coarsening never decreases loss). *If q' refines q (every q' -fiber is contained in some q -fiber), then for every q -fiber $q^{-1}(o)$, expressed as a union of q' -fibers $q'^{-1}(o'_1), \dots, q'^{-1}(o'_k)$,*

$$\mathcal{L}_q(o) \geq \max_i \mathcal{L}_{q'}(o'_i).$$

Proof. $C_\Gamma(o) = \bigcup_i C_\Gamma(o'_i)$, since the Γ -values appearing in $q^{-1}(o)$ are exactly those appearing in its constituent q' -fibers. A union of sets is at least as large as any one of them, so $|C_\Gamma(o)| \geq |C_\Gamma(o'_i)|$ for each i , and log is monotone increasing. $\square$

Remark 3.2. *Merging two fibers can only add to the set of continuation spaces an observer sees when looking at the merged fiber; it can never remove one. This is the precise sense in which coarsening a quotient is a one-way trip toward loss: refining a lossy quotient can repair the loss, by splitting a mixed fiber back into pieces that each see only one Γ -value, but no further coarsening of an already-lossy quotient can undo the damage, only add to it or leave it unchanged.*

Distinction margin as a local loss detector

Proposition 3.3 (Vanishing margin detects a redundant distinction). *For a distinction $\mathcal{D}$ and $x \in K$, $A_{\mathcal{D}}^T(x) = 0$ (Chapter 7 of Continuation Geometry) if and only if some y outside x 's $\mathcal{D}$ -cell satisfies $R_T(x) = R_T(y)$: that is, $\mathcal{D}$ places x and y in different cells even though their continuation spaces coincide at horizon T . Vanishing margin is a diagnostic for $\mathcal{D}$ being finer than $\sim_\Gamma$ at x , drawing a distinction continuation geometry does not support, not for $\mathcal{D}$ being coarser.*

Proof. Recall $A_{\mathcal{D}}^T(x) := \inf_{y \notin D(x)} d_H(R_T(x), R_T(y))$. This infimum is 0 exactly when some y outside x 's $\mathcal{D}$ -cell has $R_T(x) = R_T(y)$ (attained, under the compactness hypothesis already used in Chapter 7), which is exactly the statement that x and that y share a Γ -value at horizon T despite $\mathcal{D}$ having placed them in different cells: they are $\sim_\Gamma$ -equivalent even though $\mathcal{D}$ distinguishes them. This is over-refinement, not collapse. $\square$

Definition 3.2 (Within-cell spread). *For a $\mathcal{D}$ -cell D , $\text{diam}_\Gamma(D) := \sup_{x, y \in D} d_H(R_T(x), R_T(y))$.*

Proposition 3.4 (Zero spread is the actual collapse-safety detector). $\text{diam}_\Gamma(D) = 0$ if and only if D is contained in a single $\sim_\Gamma$ -class at horizon T , i.e. $\mathcal{D}$ introduces no loss in the sense of the fiber-loss definition earlier in this chapter, applied to D as a fiber.

Proof. $\text{diam}_\Gamma(D) = 0$ means $d_H(R_T(x), R_T(y)) = 0$ for every $x, y \in D$, i.e. $R_T(x) = R_T(y)$ for every pair in D , i.e. D lies inside one $\sim_\Gamma$ -class; this is exactly $|C_\Gamma(D)| = 1$ from the definition of fiber loss, i.e. $\mathcal{L}(D) = 0$. $\square$

Remark 3.3. This gives the distinction margin of the first book a second reading, and a corrective one. It was introduced there as a measure of how close a state sits to losing a distinction, language that invites exactly the confusion the propositions above are meant to head off. $A_D^T(x) = 0$ is not a report of quotient loss; it is a report that $\mathcal{D}$ is drawing a line continuation geometry does not, distinguishing two states with identical futures. Quotient loss, in the sense the rest of this chapter measures it, is instead the province of diam_Γ : a cell D with $\text{diam}_\Gamma(D) > 0$ is the one actually merging states with different futures, the failure mode Chapter 3 of this book was built to quantify. The two diagnostics point in opposite directions, over-refinement and under-refinement, and a single distinction $\mathcal{D}$ can exhibit either, both, or neither, independently at different points of K .

The traffic light, quantified

Return to the two designs compared in the previous chapter: the correct one, clearing on arrival at phase 0, and the alternative, clearing whenever currently at phase 0.

Example 3.1 (Zero loss, and one bit). The correct design's quotient is $\sim_\Gamma$ itself (Proposition 4 of the previous chapter), so by Theorem 3.1 its loss is $\mathcal{L}(o) = 0$ at every fiber o . The alternative design merges $(0, 0)$ and $(0, 1)$ into a single fiber o_{bad} . Their Γ -values at $T = 1$ are $R_1(0, 0) = \{(1, 0), (1, 1)\}$ and $R_1(0, 1) = \{(1, 1)\}$, which are distinct, so $C_\Gamma(o_{\text{bad}}) = \{\{(1, 0), (1, 1)\}, \{(1, 1)\}\}$, a set of size 2, giving

$$\mathcal{L}(o_{\text{bad}}) = \log 2,$$

exactly one bit, and $\mathcal{L}(o) = 0$ at every other fiber, since the alternative design agrees with the correct one everywhere except at this single merge.

Remark 3.4. This is what the previous chapter's qualitative verdict, that the alternative design was coarser and information-destroying, actually amounts to once measured: not a diffuse degradation spread across the whole system, but a single, exactly located bit of loss, sitting at precisely the fiber where the alternative design merged two states that a patron of this particular library, a controller trying to plan four steps ahead, would still have wanted to tell apart. Every other fiber of the alternative design is, by coincidence or by the limited scope of its one design error, exactly as good as the correct one.

What loss does not yet explain

This chapter can say, precisely, how much is lost by a given quotient and exactly where. It has nothing to say about why a particular coarser quotient might be chosen anyway, in full knowledge of the loss it incurs. A designer might prefer a one-bit-lossy traffic light if the saved bit of memory matters more than the distinction it would have preserved; a librarian might

prefer the cheap two-category catalog if most patrons never need the finer one. Measuring loss and deciding whether a loss is worth paying for are different questions, and this chapter has only ever answered the first of them.

Chapter 4

Continuation Operators

The first three chapters of this book have all, in their own way, been about nouns: a shape a system of contractions converges to, a partition of histories that share a future, a number measuring what a partition throws away. None of them named the verb. A contraction does something to a shape. A design decision, clearing a bit on arrival at a red light rather than while sitting at one, does something to a history. This chapter is about the doing: what counts as a legitimate move in this framework, and what survives when two different moves are composed into one.

The two kinds of move already used in this book look, on the surface, like opposites. Chapter 1's contractions built detail up, each application adding a smaller, self-similar copy of the whole. Chapters 2 and 3's quotients tore detail down, collapsing histories that had become, for every purpose this book has formalized, the same. It would be a missed opportunity to leave them looking like two unrelated tools. They are both instances of one idea, and this chapter's job is to say what that idea is.

Operators and their composition

Definition 4.1 (Continuation operator). *A continuation operator on X is a Lipschitz map $T : \mathcal{K}(X) \rightarrow \mathcal{K}(X)$: there is a constant $L \geq 0$ with $d_H(T(A), T(B)) \leq L d_H(A, B)$ for all $A, B \in \mathcal{K}(X)$. Write $L(T)$ for the smallest such constant.*

Proposition 4.1 (Closure under composition). *If T_1, T_2 are continuation operators, so is $T_2 \circ T_1$, with $L(T_2 \circ T_1) \leq L(T_2) L(T_1)$.*

Proof. $d_H(T_2(T_1(A)), T_2(T_1(B))) \leq L(T_2) d_H(T_1(A), T_1(B)) \leq L(T_2) L(T_1) d_H(A, B)$. □

Remark 4.1. *This proof is one line because it has to be; the definition was chosen precisely so that composition would cost nothing further to verify. What makes the definition worth having is not this proposition but what it is a common home for. The contractions of Chapter 1 are continuation operators with $L(T) < 1$. The map $\Psi(A) := \bigcup_{z \in A} \Phi(z)$ used throughout Chapters 2 and 3 to advance a reachable set by one step is a continuation operator too, generally with $L(\Psi) \geq 1$: it need not contract, but it is still Lipschitz whenever the underlying one-step map Φ is well behaved. Both belong to the same category. What distinguishes a generative operator from a reductive one is not the definition above; it is what happens under repeated composition, contraction toward a single attractor in one case, and, as the next section makes precise, collapse of distinctions in the other.*

Operators on histories, and what they respect

The operators above act on shapes directly. A different and equally natural kind of operator acts on histories: appending one more real transition, editing a codebase, applying a repair. The interesting question for an operator of this kind is not its Lipschitz constant but whether it interacts predictably with $\sim_\Gamma$.

Definition 4.2 (Γ -compatible operator). *$T : H \rightarrow H$ is Γ -compatible if there is a map $\hat{T} : \mathcal{K}(X) \rightarrow \mathcal{K}(X)$ with $\Gamma(T(h)) = \hat{T}(\Gamma(h))$ for every $h \in H$: the continuation space of the transformed history depends on the original history only through its continuation space, not otherwise.*

Proposition 4.2 (Γ -compatible operators descend to the quotient). *If T is Γ -compatible, then $h_1 \sim_\Gamma h_2 \implies T(h_1) \sim_\Gamma T(h_2)$: T induces a well-defined map $\bar{T} : H/\sim_\Gamma \rightarrow H/\sim_\Gamma$.*

Proof. This is Proposition 4 of the previous chapter (quotient invariance), applied to $F := \hat{T} \circ \Gamma$ in place of Γ itself: $h_1 \sim_\Gamma h_2$ gives $\Gamma(h_1) = \Gamma(h_2)$, hence $\Gamma(T(h_1)) = \hat{T}(\Gamma(h_1)) = \hat{T}(\Gamma(h_2)) = \Gamma(T(h_2))$, so $T(h_1) \sim_\Gamma T(h_2)$. $\square$

Remark 4.2. *This is the sense in which Γ -compatibility is the right condition to ask for: it says an operator cannot secretly depend on information that continuation equivalence has already declared irrelevant. An operator failing this condition would be able to treat two continuation-equivalent histories differently for reasons that have nothing to do with what either history can still become, which is exactly the kind of dependence Chapter 5 of Continuation Geometry worried about under the name of history-dependence in the first place.*

Example 4.1 (The traffic light's step operators). *For $e \in \{0, 1\}$, let $T_e : H \rightarrow H$ append the transition $\Phi(\cdot, e)$ to a history. Whether T_e is Γ -compatible (at horizon $T = 1$) needs checking only on the one nontrivial $\sim_\Gamma$ -class found in Chapter 2, $\{(3, 0), (3, 1)\}$, since every other class is a singleton and compatibility there is automatic. Direct computation gives $T_0(3, 0) = T_0(3, 1) = (0, 0)$ and $T_1(3, 0) = T_1(3, 1) = (0, 0)$: both step operators send the two equivalent histories to the identical resulting state, not merely to states with equal R_1 , so compatibility holds here in the strongest possible form.*

Remark 4.3. *This is a narrow check, confined to the single class where there was anything to check, and it should not be read as a general fact about T_0 or T_1 on this system, only as a confirmation that neither step operator introduces a new dependence on which of $(3, 0)$ or $(3, 1)$ a history happened to pass through, at exactly the point where such a dependence would have mattered.*

The family Γ_T and the operator Ψ

A different relationship, easy to conflate with Γ -compatibility but distinct from it, has been used without a name since Chapter 2. It concerns not an operator on histories but the relationship between the finite-horizon continuation spaces at consecutive horizons for one and the same history.

Remark 4.4 (Ψ relates horizons, not histories). *By construction, $R_{T+1}(x) = \bigcup_{z \in R_T(x)} \Phi(z) = \Psi(R_T(x))$, i.e. $\Gamma_{T+1} = \Psi \circ \Gamma_T$ as an identity between functions $H \rightarrow \mathcal{K}(X)$, where $\Gamma_T(h) := R_T(x(h))$. This is not an instance of Proposition 4.2; no operator on H is involved, only the fixed*

continuation operator Ψ acting on the shape side, once for each increment of the horizon. It is this identity, not any property of an operator on histories, that did the real work in Lemma 2 of the previous chapter (one-step agreement propagates) and in the eventual-periodicity argument of Chapter 3: both are, in retrospect, statements about the orbit of a single reachable set under repeated application of the one continuation operator Ψ , dressed in the language of horizons rather than the language of operators used here.

Two families, one category

Corollary 4.1. *Both the contractions of Chapter 1 and the operator Ψ of Chapters 2 and 3 are continuation operators in the sense of this chapter's first definition; they differ only in whether their Lipschitz constant is below 1.*

Remark 4.5. *It is tempting to read this as saying generation and collapse are two regions of one parameter, contraction below 1, something else above it, and that reading is too strong. $L(T) < 1$ does imply convergence to a unique attractor, by Chapter 1's own theorem; that direction is solid. $L(T) \geq 1$ implies nothing in particular. The identity map has $L = 1$ and collapses nothing. An isometry has $L = 1$ and collapses nothing. A map that stretches every distance by a factor of two has $L = 2$ and, again, collapses nothing, since stretching does not identify points that were previously distinct. What actually collapses distinctions is a property orthogonal to the Lipschitz constant: whether the operator is injective, whether distinct inputs can share an output. Ψ collapses states in the traffic light not because $L(\Psi) \geq 1$, but because the specific map Φ underlying it is non-injective at the clearing transition, sending both $(3, 0)$ and $(3, 1)$ to $(0, 0)$; a different one-step map with $L(\Psi) \geq 1$ but no collisions anywhere would generate no quotient at all. The Lipschitz constant and injectivity are independent properties an operator can carry in any combination, and conflating them is exactly the kind of overstatement this book has tried, elsewhere, to catch in itself before it went to print.*

Remark 4.6 (A more honest organizing geometry). *What Chapters 1 through 3 actually populate is not a line but a small space of independent properties: metric sensitivity ($L(T)$, whether and how strongly the operator contracts or expands distances); injectivity (whether the operator identifies distinct states, independent of $L(T)$); and compatibility with Γ (Chapter 4's own Γ -compatibility condition, a further independent property an operator either has or lacks). A contraction with $L < 1$ and full injectivity, the pure case of Chapter 1, generates a self-similar attractor without collapsing anything along the way. A non-injective Ψ with $L(\Psi) \geq 1$, the traffic light's case, collapses distinctions and does not converge to a point, cycling instead. Other combinations, injective maps with $L \geq 1$, non-injective contractions, are equally coherent and simply were not this book's running examples. The genuine unification this chapter earned is that all of these live in one vocabulary, continuation operators, checkable by the same closure proposition regardless of where in this space they sit. The overclaim to retract is that a single number tells you where in that space an operator is.*

What operators do not yet explain

This chapter has said what a legitimate move is and shown that both of this book's central constructions are instances of it, but it has not asked which finite collections of operators are rich enough to produce every continuation geometry worth having, only that individual operators compose predictably once they exist. That question, closer to the universal-generation

intuition this project began with, needs the vocabulary built here as a prerequisite, and is not attempted in this chapter.

Chapter 5

Toward Universal Generation

A handful of letters generates every word in a dictionary that has not been written yet. A handful of chess rules generates every game that will ever be played under them, including games no one has conceived of. A small, fixed vocabulary reaching an unbounded space of outcomes is one of the oldest tricks available to any generative system, and it is worth asking, now that this book has a settled vocabulary for what a continuation operator is, whether continuation geometries have the same property. Can a small, fixed family of operators, chosen once and never enlarged, get arbitrarily close to any admissible continuation geometry at all?

This chapter does not answer that question. It draws its boundary precisely enough to say exactly what would need to be true for the answer to be yes, proves the one direction that is actually within reach, and is honest about the direction that is not. The gap between those two is real mathematics, not a gap in effort, and closing it is left where it belongs: open.

The conjecture, stated precisely

Conjecture 5.1 (Universal continuation generation). *There exists a finite family of point contractions $T_1, \dots, T_n : X \rightarrow X$ and a single seed set $P \in \mathcal{K}(X)$, both fixed once and independent of any particular target, such that for every nonempty compact $A \subseteq X$ and every $\varepsilon > 0$, some finite word $i_1, \dots, i_k$ satisfies*

$$d_H(A, (T_{i_k} \circ \dots \circ T_{i_1})(P)) < \varepsilon,$$

where $(T_{i_k} \circ \dots \circ T_{i_1})(P) := \{(T_{i_k} \circ \dots \circ T_{i_1})(p) : p \in P\}$ is the composed point-map, itself a single contraction, applied setwise to the fixed seed.

Remark 5.1 (Why the seed must be fixed). *Allowing the starting set to vary with A would make the conjecture trivial: given any target A and any ε , take the starting set to be an ε -net of A itself, apply the identity word, and the approximation is immediate regardless of what the T_i are. The entire content of the conjecture is that one seed and one generator family, chosen before any target is known, must work for every target afterward; fixing P once, alongside the T_i , is not an optional refinement of the statement, it is the statement.*

Remark 5.2. *This is stated as a conjecture and not as a theorem because it is one; nothing in this book has proved it, and nothing in this chapter will. What can be made precise is why it is hard, and that requires first showing what is not hard.*

The Collage Theorem

Theorem 5.1 (Collage Theorem). *Let $T_1, \dots, T_n$ be contractions with $\mathcal{H}(A) = \bigcup_i T_i(A)$ and maximum contraction constant $c < 1$, and let A^* be the attractor of $\mathcal{H}$. Then for any nonempty compact A ,*

$$d_H(A, A^*) \leq \frac{d_H(A, \mathcal{H}(A))}{1 - c}.$$

Proof. By the triangle inequality and $A^* = \mathcal{H}(A^*)$,

$$d_H(A, A^*) \leq d_H(A, \mathcal{H}(A)) + d_H(\mathcal{H}(A), \mathcal{H}(A^*)).$$

By Proposition 3 of Chapter 1 ($\mathcal{H}$ is a contraction on shapes), $d_H(\mathcal{H}(A), \mathcal{H}(A^*)) \leq c d_H(A, A^*)$. Substituting,

$$d_H(A, A^*) \leq d_H(A, \mathcal{H}(A)) + c d_H(A, A^*),$$

and rearranging, since $c < 1$, gives $(1 - c) d_H(A, A^*) \leq d_H(A, \mathcal{H}(A))$, which is the claim. $\square$

Remark 5.3. *Nothing here was invented for this chapter; the theorem is a direct three-line consequence of facts Chapter 1 already established, and it is worth noting how much it buys for how little it costs. It converts a question about the attractor, an object defined only as a limit, so generally hard to reason about directly, into a question about a single application of $\mathcal{H}$ to a set already in hand. Finding an operator family whose attractor is close to a target A reduces to finding one whose one-step collage $\mathcal{H}(A)$ is already close to A , an ordinary geometric fitting problem rather than a limiting one.*

Existence of an approximating family

Theorem 5.2 (Every compact target is approximable). *For every nonempty compact $A \subseteq X = \mathbb{R}^n$ and every $\varepsilon > 0$, there exists a finite family of contractions (depending on A and ε) whose attractor A^* satisfies $d_H(A, A^*) < \varepsilon$.*

Proof. Let $\{x_1, \dots, x_m\} \subseteq A$ be a finite δ -net of A with $\delta < \varepsilon$, which exists since A is compact, hence totally bounded, and can be chosen with net points drawn from A itself. For each i , let $T_i : X \rightarrow X$ be the constant map $T_i(x) := x_i$. Each T_i is trivially a contraction, with constant $c = 0$, since $d(T_i(x), T_i(y)) = 0 \leq 0 \cdot d(x, y)$ for all x, y . Then $T_i(A) = \{x_i\}$ exactly, for any nonempty A , so $\mathcal{H}(A) = \bigcup_i T_i(A) = \{x_1, \dots, x_m\}$, the net itself. Since the net points lie in A and every point of A is within δ of the net, $d_H(A, \mathcal{H}(A)) = d_H(A, \{x_1, \dots, x_m\}) \leq \delta$. By Theorem 5.1 with $c = 0$, $d_H(A, A^*) \leq d_H(A, \mathcal{H}(A))/(1 - 0) \leq \delta < \varepsilon$. $\square$

Remark 5.4. *This proof is fully elementary once constant maps are admitted as contractions, which they trivially are under the definition used throughout this book, and it is worth saying so plainly rather than gesturing at a more elaborate construction that turns out not to be needed. A version restricted to contractions with ratio bounded away from 0 demands more care, positioning each T_i so its image lands near x_i without collapsing to a point, and is the version classically used in fractal image compression [3]; that version is not reproduced here, since the elementary one already proves everything this theorem claims.*

Remark 5.5. *This is real progress, and it is worth being precise about exactly how much. It says the space of Hutchinson attractors is dense, in Hausdorff distance, in the space of nonempty compact subsets of X : nothing worth generating is permanently out of reach. What it does not say, and what Conjecture 5.1 actually asks for, is that a single family, fixed before any target is known, suffices for every target. The construction above produces a different family for each new target and each new tolerance; as $\varepsilon \rightarrow 0$ or as A grows more intricate, the number of generators m used in the proof generally grows without bound. The existence theorem answers “can this be reached at all,” which Theorem 5.2 settles completely. The conjecture asks “can it be reached with tools chosen in advance,” which is a categorically stronger demand, and this chapter has not moved that needle.*

What kind of gap this is

The distinction is not a technicality, and it has a name in neighboring fields worth being explicit about rather than gestured at. The Stone–Weierstrass theorem [5] guarantees that polynomials are dense in the continuous functions on a compact interval: a fixed class of building blocks, closed under the operations that generate it, dense in everything worth approximating. A universal approximation theorem for neural networks [6] says something structurally similar: networks built from a fixed activation function, with enough width, are dense in the continuous functions. Both are theorems about a fixed generator class being rich enough for every target, which is exactly the shape of Conjecture 5.1 and exactly what Theorem 5.2 does not provide, since its generator family is chosen after the target is known.

Remark 5.6. *Whether an analogue of Stone–Weierstrass or universal approximation holds for continuation geometries specifically is not something this chapter’s methods can settle, and it should not be assumed to transfer just because the surface shape of the claim resembles those theorems. Both classical results rely on structure, algebraic closure under multiplication for Stone–Weierstrass, compositional depth for neural networks, that has no established analogue here. Building that structure, or showing it cannot exist, is what would actually resolve Conjecture 5.1, and it remains the open question this book set out to sharpen rather than close.*

Chapter 6

State Insufficiency, Made Explicit

A doctor at the bedside observes breathing, heart rate, pupil response, the ordinary vital signs a chart records every hour. Two patients can match on every one of them, one sleeping, one comatose, and the chart alone will not say which. This is not a failure of the instruments. It is a fact about what the instruments were built to see. The chart records an observation, a coarse summary of a much richer state, and coarse summaries are allowed to erase exactly the distinctions that happen not to show up in whatever they were designed to measure. Whether a given summary erases a distinction that matters, mere resemblance in the readings or a genuine difference in what each patient can still become, is not a question the chart can answer about itself.

This idea has been the motivating image behind this project since before either book in it had a name, and it has never been made into a theorem. It does not need a new toy system to become one. The traffic light, already built and already verified twice over in Chapters 2 and 3, contains a clean instance of exactly this phenomenon, once it is looked at through a coarser observation than the one this book has been using.

Observations and sufficiency

Definition 6.1 (Observation map, state sufficiency). *An observation map is any $\Pi : X \rightarrow O$. Π is state-sufficient (for continuation, at horizon T) if $\Pi(x) = \Pi(y) \implies R_T(x) = R_T(y)$ for all $x, y \in X$; otherwise Π is state-insufficient.*

Remark 6.1. *This is deliberately the same shape of question Chapter 3 already answered in general. Π is a quotient of X in exactly the sense used throughout this book, and state sufficiency is exactly zero loss in the sense of that chapter, transported from quotients of histories to observation maps on states. The next proposition makes the translation exact rather than merely suggestive.*

Proposition 6.1 (Sufficiency is refinement of $\sim_\Gamma$). *Π is state-sufficient at horizon T if and only if Π refines $\sim_\Gamma$ at that horizon: $\Pi(x) = \Pi(y) \implies x \sim_\Gamma y$.*

Proof. Immediate from the definitions: $x \sim_\Gamma y$ at horizon T means $R_T(x) = R_T(y)$, which is exactly the consequent of state sufficiency's defining implication. $\square$

An explicit failure of sufficiency

Definition 6.2. Let $\Pi : \{0, 1, 2, 3\} \times \{0, 1\} \rightarrow \{0, 1, 2, 3\}$, $\Pi(k, r) := k$, the observation that records the traffic light's phase but not its pending-request bit.

Theorem 6.1 (State insufficiency, explicit instance). Π is state-insufficient at $T = 1$: $\Pi(0, 0) = \Pi(0, 1) = 0$, but $R_1(0, 0) = \{(1, 0), (1, 1)\} \neq \{(1, 1)\} = R_1(0, 1)$.

Proof. Both values of Π are immediate from the definition of Π . The inequality of the reachable sets is the computation already carried out in Chapter 2 (Proposition 3, the $T = 1$ traffic-light computation): $R_1(0, 0)$ and $R_1(0, 1)$ were shown there to differ, as one of the seven singleton classes and part of the argument establishing that no merge occurs at phase 0. $\square$

Remark 6.2. This is the sleep/coma image with every part of it named. Π plays the role of the chart: it records the phase, the part of the state a casual glance would call “where things currently stand,” and discards the request bit, the part recording something about how the state got there and what it is still owed. Two states indistinguishable under Π , $(0, 0)$ and $(0, 1)$, are not indistinguishable under Γ : one of them can still become $(1, 0)$, a future the other cannot reach at all. The chart is not lying. It was simply never built to record the fact that would have mattered here.

How much sufficiency fails, and where

Chapter 3 built exactly the machinery needed to go beyond a single witnessing pair and measure the full extent of Π 's insufficiency.

Proposition 6.2 (The loss of Π , computed exactly). For each phase $k \in \{0, 1, 2, 3\}$, the loss at fiber $\Pi^{-1}(k) = \{(k, 0), (k, 1)\}$ is

$$\mathcal{L}(\Pi^{-1}(k)) = \begin{cases} \log 2 & k \in \{0, 1, 2\} \\ 0 & k = 3. \end{cases}$$

Proof. By definition, $\mathcal{L}(\Pi^{-1}(k)) = \log |\{R_1(k, 0), R_1(k, 1)\}|$. From the values already established (Chapter 2, Proposition 3, and Theorem 6.1 above): $R_1(0, 0) \neq R_1(0, 1)$, $R_1(1, 0) = \{(2, 0), (2, 1)\} \neq \{(2, 1)\} = R_1(1, 1)$, and $R_1(2, 0) = \{(3, 0), (3, 1)\} \neq \{(3, 1)\} = R_1(2, 1)$, giving $|\{\cdot, \cdot\}| = 2$ and loss $\log 2$ in each case. At $k = 3$, $R_1(3, 0) = R_1(3, 1) = \{(0, 0)\}$, giving $|\{\cdot\}| = 1$ and loss 0. $\square$

Remark 6.3. The pattern is worth reading rather than only computing. The observation Π loses exactly one bit at every phase except the one immediately before clearing, where it loses nothing, for exactly the reason Chapter 3's eventual-redundancy result already identified: at phase 3, the request bit is one step from becoming irrelevant regardless of what Π does, so discarding it early costs nothing there and costs a full bit everywhere else. State insufficiency, once quantified rather than merely demonstrated, is not a uniform fog sitting over the whole system; it has a precise shape, concentrated exactly where the discarded information still had work left to do.

What this does and does not establish about sleep and coma

Nothing above is a claim about physiology. It is a claim about one eight-state machine and one particular coarsening of it, chosen because the numbers needed to state it precisely were already on hand and already checked. Whether any actual clinical observation, a chart, a scan, a bedside exam, is state-insufficient for the specific continuation spaces that distinguish sleep from coma is an empirical question about biology, not something a toy traffic light can settle by analogy. What this chapter has shown is narrower and more useful than a claim about medicine: that state insufficiency is a real, precisely statable, and exactly measurable property of an observation map relative to a dynamics, that it can fail in some places and not others within the very same system, and that the machinery already built in this book, with no further construction, is already enough to find out which.

Chapter 7

Invariant Decomposition, Staleness, and Intervention

A scouting report can be wrong in two entirely different ways. It can be out of date: accurate when filed, and steadily less trustworthy with every day that passes since, repairable by sending another scout. Or it can be asking the wrong question entirely: a report on one trading region says nothing, however fresh, about a second region no ship has ever crossed into, because the two have never exchanged a single price, a single rumor, a single unit of anything. No amount of resurveying the first region brings the second any closer to being known. These look like the same complaint, an observer failing to keep up with reality, and they are not. One is a fact about elapsed time. The other is a fact about the shape of the space itself. This chapter gives each its own formal treatment, because giving them the same one would hide the fact that they need different remedies.

Invariant decomposition

Definition 7.1 (Invariant block). *$A \subseteq X$ is an invariant block under F if $R_1(x) \subseteq A$ for every $x \in A$.*

Proposition 7.1 (Invariance propagates). *If A is an invariant block, then $R_T(x) \subseteq A$ for every $x \in A$ and every $T \geq 0$.*

Proof. By induction on T . $T = 0$ is immediate ($x \in A$). If $R_T(x) \subseteq A$, then $R_{T+1}(x) = \bigcup_{z \in R_T(x)} R_1(z) \subseteq \bigcup_{z \in A} R_1(z) \subseteq A$, the last inclusion by the definition of an invariant block applied to each $z \in A$. $\square$

Definition 7.2 (Invariant decomposition). *X admits an invariant decomposition into $\{X_i\}$ if the X_i partition X and each X_i is an invariant block.*

Corollary 7.1 (Continuation confinement). *If $x \in X_i$, then $\Gamma_K(x) \subseteq \{\gamma : \gamma(t) \in X_i \forall t\}$: every admissible continuation of x remains in X_i forever.*

Proof. Immediate from Proposition 7.1: $R_T(x) \subseteq X_i$ for every T , and a trajectory whose position at every finite horizon lies in X_i lies in X_i at every time. $\square$

Remark 7.1. *This is the precise sense in which observation in one block cannot substitute for observation in another. For $x \in X_1$, $\Gamma_K(x)$ is determined entirely by the dynamics restricted to X_1 ; nothing about X_2 , however thoroughly sampled, enters into $R_T(x)$ for any T , because Corollary 7.1 rules out any trajectory from x ever visiting X_2 in the first place. This is not a claim that X_2 is hard to observe. It is a claim that X_2 is, formally, irrelevant to x 's continuation space once the decomposition holds, in the same sense that a variable a function does not depend on is irrelevant to computing it, however precisely that variable is measured. The classical distinction between a viability kernel (admitting some trajectory that stays put) and an invariance kernel (requiring every trajectory to stay put), already present in the literature this book has cited since Chapter 1 [7], is exactly the distinction an invariant block relies on, and exactly why this section is named for invariance rather than for ergodicity: nothing here involves a measure or an indecomposability condition on invariant measures, the actual content of the word ergodic in its established sense. What is built here is the purely topological half of that classical picture.*

Staleness

Invariant decomposition is a permanent, structural fact about a system. Staleness is not: it is what happens to an observer's knowledge of a single, well-connected system as time passes without a fresh look.

Definition 7.3 (Staleness bound). *Suppose $\dot{x} = f(x, u(t))$ for a fixed perturbation space Q (independent of x) and $u(t) \in Q$ measurable, with f Lipschitz in x with constant L uniformly in $u \in Q$: $\|f(x, u) - f(y, u)\| \leq L\|x - y\|$ for all $u \in Q$. Given an observation fiber $\Pi^{-1}(o)$ of diameter $d_0 := \sup_{x, y \in \Pi^{-1}(o)} d(x, y)$, the staleness bound at elapsed time Δt is $d_0 e^{L\Delta t}$.*

Proposition 7.2 (Staleness grows at most exponentially). *For $x, y \in \Pi^{-1}(o)$ and trajectories γ_x, γ_y solving $\dot{\gamma}_x = f(\gamma_x, u(t))$, $\dot{\gamma}_y = f(\gamma_y, u(t))$ from x, y under the same measurable input $u(t) \in Q$, $d(\gamma_x(\Delta t), \gamma_y(\Delta t)) \leq d_0 e^{L\Delta t}$.*

Proof. Let $w(t) := d(\gamma_x(t), \gamma_y(t))$. Since f is L -Lipschitz in its first argument uniformly in $u \in Q$ and both trajectories are driven by the same $u(t)$, $\dot{w}(t) \leq \|f(\gamma_x(t), u(t)) - f(\gamma_y(t), u(t))\| \leq L w(t)$, so $w(t) \leq w(0) + L \int_0^t w(s) ds$. By the continuous Grönwall inequality (Appendix C of Continuation Geometry), $w(t) \leq w(0) e^{Lt} = d_0 e^{Lt}$, giving the claim at $t = \Delta t$. $\square$

Remark 7.2. *This is stated for single-valued dynamics with a shared perturbation space, deliberately narrower than the set-valued F used elsewhere in this book. A genuine differential inclusion, where the two trajectories are not assumed to realize a common measurable selection, needs more than a Hausdorff-Lipschitz bound on F to license the same comparison estimate; it needs an explicit coupling, typically supplied by a selection theorem of the kind Appendix B of Continuation Geometry already cites, or by a viability-theoretic argument constructing matched trajectories directly. That extension is not carried out here.*

Remark 7.3. *Nothing in this proof is new; it is the same Grönwall estimate already used to bound trajectory growth under linear growth hypotheses, applied here to the difference between two trajectories instead of the norm of one. What makes it worth restating in this form is the reading it supports: staleness is not a fixed penalty for not having observed recently, it is a bound that can always be reset. Taking a fresh observation at time Δt collapses the relevant fiber back down to whatever its new diameter is,*

and the exponential clock restarts from there. This is the formal content behind “staleness is repaired by resampling”: the bound in Proposition 7.2 is a function of elapsed time since the last observation, not of any permanent property of the system.

Remark 7.4 (Why the two do not reduce to one another). *A system separated into invariant blocks can be resampled as often as an observer likes without ever closing the gap between blocks, since no finite or infinite sequence of observations confined to X_1 changes $\Gamma_K(x)$ for $x \in X_1$, by Corollary 7.1. A stale observation of a well-connected system closes its own gap with a single fresh sample. Treating both under one heading, representation degrading relative to territory, obscures that one is fixed by waiting and looking again and the other cannot be fixed by looking again at all, only by being, or gaining access to, the other block.*

Projection and intervention

Chapter 6 formalized when an observation Π fails to determine a state’s continuation space. It said nothing about what, if anything, is done to a state once observed. The two are logically separate questions.

Definition 7.4 (Intervention operator). *An operator T (Chapter 4) is an intervention on a set $S \subseteq X$ if there is a fixed target $A_0 \in \mathcal{K}(X)$ with $\Gamma(T(x)) = A_0$ for every $x \in S$, regardless of $\Gamma(x)$.*

Proposition 7.3 (Sufficiency and intervention are independent). *There exist systems in which Π is state-sufficient on S yet an intervention operator is applied on S anyway, and systems in which Π is state-insufficient on S with no intervention applied at all.*

Proof. For the first: take Π to be the identity (trivially sufficient, since $\Pi(x) = \Pi(y) \implies x = y \implies \Gamma(x) = \Gamma(y)$), and let T map every $x \in S$ to a fixed state x_0 regardless of x ; T is an intervention by construction, and Π ’s sufficiency is unaffected by whether T is applied, since sufficiency is a property of Π alone. For the second: the traffic light’s phase-only Π of Chapter 6 is state-insufficient on $\{(0, 0), (0, 1)\}$, and the system’s actual dynamics Φ is not an intervention in the sense above, since $\Gamma(\Phi(0, 0)) \neq \Gamma(\Phi(0, 1))$ in general (the two states are not forced to a common target). $\square$

Remark 7.5. *This is a narrow, formal point, and it is worth being honest about how narrow: it shows the two properties are not logically entailed by one another, not that they are uncorrelated in how real institutions actually behave. An institution whose observation is coarse may find uniform treatment easier to justify or administer than individually-tailored treatment would be, which is a pressure toward intervention that has nothing to do with logical necessity and everything to do with what coarse observation makes practical. The independence proposition above rules out the claim that better observation requires less intervention or that worse observation requires more; it says nothing about which correlations actually hold in systems built by people operating under those practical pressures, which is a separate, empirical question this chapter’s methods do not reach.*

What these three have in common

Ergodic separation, staleness, and intervention are three distinct ways a system’s continuation geometry can be misjudged, by structure, by elapsed time, and by force, and this chapter has

kept them formally apart on purpose. Collapsing them into one diagnosis, an observer failing to track reality, would have been easier to write and less useful to read: each has its own signature, its own proof, and its own remedy, and only one of the three is fixed by looking again.

Chapter 8

Investment-Movable Admissibility

Not every boundary asks the same thing of you. A species cannot breathe an atmosphere its lungs did not evolve for, and no amount of committed effort changes that fact; the boundary is simply where it is, permanently, for reasons that have nothing to do with anyone's budget. A planet's atmosphere, on the other hand, can be thickened, warmed, seeded, and made breathable by exactly the kind of committed effort the first boundary is indifferent to. Both are limits on what is admissible. Only one of them recedes when you push on it.

Every constraint set K used so far in this book and its predecessor has been fixed once and for all, a single unmoving boundary between what a system may become and what it may not. This chapter asks what changes when that boundary is allowed to move, not arbitrarily, but in response to something accumulated over time, and does so using exactly the operator vocabulary Chapter 4 already built, applied to a kind of object that chapter did not yet ask it to touch.

Constraint sets are shapes too

Chapter 4 defined a continuation operator as a Lipschitz map $T : \mathcal{K}(X) \rightarrow \mathcal{K}(X)$ and used it exclusively to transform reachable sets. Nothing in that definition mentions reachability. A constraint set K , provided it is compact, is simply another element of $\mathcal{K}(X)$, and there is nothing preventing an operator from acting on it directly.

Definition 8.1 (Investment family). *An investment family is a collection of continuation operators $\{E_\delta\}_{\delta \geq 0}$ on $\mathcal{K}(X)$ satisfying:*

- (i) $E_0 = \text{id}$;
- (ii) $E_{\delta_1 + \delta_2} = E_{\delta_2} \circ E_{\delta_1}$ for all $\delta_1, \delta_2 \geq 0$ (a one-parameter semigroup);
- (iii) $K \subseteq E_\delta(K)$ for every $K \in \mathcal{K}(X)$ and $\delta \geq 0$ (expansive).

Given a base constraint set K_0 , the investment schedule is $K_i := E_i(K_0)$ for accumulated investment $i \geq 0$.

Proposition 8.1 (The schedule is monotone). $K_i \subseteq K_j$ whenever $i \leq j$.

Proof. Write $j = i + \delta$ for $\delta = j - i \geq 0$. By the semigroup property, $K_j = E_j(K_0) = E_\delta(E_i(K_0)) = E_\delta(K_i)$, and by expansiveness, $K_i \subseteq E_\delta(K_i) = K_j$. $\square$

Remark 8.1. This is the formal content of “investment never shrinks admissibility”: once the family satisfies the three axioms above, monotonicity is not an additional assumption, it is forced. What the axioms do not force, and what distinguishes one investment family from another, is how much room a given increment of investment actually buys, which is exactly the content of the operators E_δ themselves rather than of the schedule they generate.

Hard and movable boundaries

Definition 8.2 (Hard-excluded set). $K_\infty := \overline{\bigcup_{i \geq 0} K_i}$, the closure of the investment orbit. A state $x \in X$ is hard-excluded if $x \notin K_\infty$; otherwise x is investment-reachable.

Remark 8.2. The closure is necessary, not decorative: a monotone union of compact sets need not itself be compact, or even closed, since the union can approach a boundary point without ever reaching it. K_∞ as defined here is the smallest closed set containing every stage of the schedule, which coincides with the plain union exactly when that union happens to be closed already, and not otherwise.

Remark 8.3. This makes the opening distinction exact. A species trait sits outside K_∞ for every conceivable investment family a design could choose, not merely the one at hand; a planet’s habitability threshold sits inside K_i for i large enough, under an investment family that models what terraforming can actually do. Whether a given boundary is hard or movable is not a property of the boundary alone; it is a property of the boundary together with whichever investment family is available to act on it, and the same physical limit can be hard under one family and movable under a richer one.

Proposition 8.2 (Viability is upward-closed in investment). If $x \in \text{Viab}_F(K_i)$ for some i , then $x \in \text{Viab}_F(K_j)$ for every $j \geq i$.

Proof. $x \in \text{Viab}_F(K_i)$ gives a trajectory $\gamma \in \Gamma_{K_i}(x)$ remaining in K_i for all time. By Proposition 8.1, $K_i \subseteq K_j$, so γ remains in K_j for all time as well, giving $\gamma \in \Gamma_{K_j}(x)$ and hence $x \in \text{Viab}_F(K_j)$. $\square$

Remark 8.4. Once a state becomes viable at some investment level, no further investment can undo that; the set of investment levels at which a given state is viable, if nonempty, is upward-closed. This is a weak fact stated for a reason: it is exactly strong enough to justify treating investment as a one-way ratchet on viability and no stronger, since Proposition 8.2 says nothing about whether a state becomes viable at any finite level in the first place, only that viability, once reached, is never lost to further investment.

A worked schedule

Example 8.1 (Habitability threshold). Let $X = [0, 1]$ represent a planet’s habitability score, and let $\tau(i) := \max(\tau_{\min}, \tau_0 - ci)$ for constants $\tau_0 \in (0, 1)$, $c > 0$, and a floor $\tau_{\min} \in [0, \tau_0)$ representing an absolute physical limit, pressure or temperature bounds no terraforming budget can cross. Define $K_i := [\tau(i), 1]$.

Proposition 8.3. This schedule arises from a valid investment family, and $K_\infty = [\tau_{\min}, 1]$, already closed.

Proof. Define E_δ on all of $\mathcal{K}(X)$ by the dilation $E_\delta(K) := \{x \in [0, 1] : \text{dist}(x, K) \leq c\delta\} \cap [\tau_{\min}, 1]$, capping at the hard floor. On $[0, 1] \subset \mathbb{R}$ with the standard metric, dilation composes exactly, $E_{\delta_2}(E_{\delta_1}(K)) = E_{\delta_1 + \delta_2}(K)$, since a point within $c\delta_2$ of a point within $c\delta_1$ of K is within $c(\delta_1 + \delta_2)$ of K by the triangle inequality, and conversely, giving the semigroup property; $K \subseteq E_\delta(K)$ is immediate, giving expansiveness; and E_δ is Lipschitz in Hausdorff distance with constant 1, a standard property of metric dilations. Restricted to the orbit generated from $K_0 = [\tau_0, 1]$, $E_\delta([\tau(i), 1]) = [\max(\tau_{\min}, \tau(i) - c\delta), 1] = [\tau(i + \delta), 1]$, matching the schedule's definition exactly. As $i \rightarrow \infty$, $\tau(i)$ reaches $\tau_{\min}$ *exactly* once $i \geq i_0 := (\tau_0 - \tau_{\min})/c$, not merely in the limit, so $K_i = [\tau_{\min}, 1]$ for all $i \geq i_0$, and $\bigcup_i [\tau(i), 1] = [\tau_{\min}, 1]$ is already closed: for this particular schedule, the closure in the definition of K_∞ adds nothing, though it would be needed for a schedule approaching its floor only asymptotically. $\square$

Remark 8.5. *Everything below $\tau_{\min}$ is hard-excluded regardless of budget; everything in $[\tau_{\min}, \tau_0]$ is investment-reachable but not part of the original admissible set, exactly the region terraforming is supposed to buy; and $\tau_{\min}$ itself is where the two kinds of boundary this chapter opened with meet, the point past which no further increment of E_δ does anything at all, since τ is already at its floor.*

An open question

Open Problem 8.1. *Chapter 1 showed that a family of contracting operators drives any starting shape to a unique attractor from outside, in the sense that repeated application converges in Hausdorff distance regardless of where it starts. An investment family is expansive rather than contracting, but it is monotone and, in examples like the one above, bounded above by a fixed K^{hard} . Whether a bounded, monotone, expansive operator family converges to a well-defined limit shape in an analogous sense, a kind of attractor approached from below rather than surrounded from outside, is not addressed here. The habitability example converges to $[\tau_{\min}, 1]$ by direct computation, but nothing in this chapter establishes that this always happens, only that it happened once.*

Closing

Nothing about the operators in this chapter required a new definition; Chapter 4's continuation operators, applied to constraint sets instead of reachable sets, were already enough. What was new was the question, not the machinery, and that is the right way around for a book that has tried, throughout, to reuse what it already has before reaching for something else.

Chapter 9

Semantic Manifolds

Call a light red and you have already thrown away almost everything true about it: which fraction of a second into the cycle it is, how many pedestrians are waiting, whether a request was just registered or has been pending for a while. None of that survives the word. And yet the word is not a failure of description, it is the entire point of having one. A category is supposed to discard detail. The question this chapter actually has to answer is not whether “red” throws things away, every category does, but whether there is a principled mathematical object underneath the word at all, or whether “meaning” was always going to be a modeling choice dressed up as a discovery.

An earlier sketch of this project wanted to say that a semantic category is an equivalence class of histories under $\Gamma(h_1) \cong \Gamma(h_2)$, and stopped there, with $\cong$ never specified. That is not a theorem with a gap in it. It is a sentence that has not yet decided what it is about. This chapter decides, and part of deciding honestly is admitting that the most obvious candidate is wrong before proposing a better one.

The obvious candidate is too strict

The most literal reading of $\Gamma(h_1) \cong \Gamma(h_2)$ available in this book is equality, $\Gamma(h_1) = \Gamma(h_2)$, which is exactly $\sim_\Gamma$ from Chapter 2. Nothing new would be needed to state a theorem this way. Something would be lost by doing so.

Proposition 9.1. *Ordinary semantic categories are, in general, coarser than $\sim_\Gamma$: there exist h_1, h_2 commonly called by the same name with $\Gamma(h_1) \neq \Gamma(h_2)$.*

Proof. Chapter 6, Theorem 1: $(0, 0)$ and $(0, 1)$, both ordinarily described as “waiting at the red light,” satisfy $R_1(0, 0) = \{(1, 0), (1, 1)\} \neq \{(1, 1)\} = R_1(0, 1)$. $\square$

Remark 9.1. *This was already proved for a different purpose, and it settles the question immediately: if $\cong$ is read as literal equality, “red light” is not a semantic category in this book’s sense at all, since two states any ordinary speaker would call by that name already fail to be $\sim_\Gamma$ -equivalent. Meaning, whatever it turns out to be, cannot be this strict without contradicting how the word is actually used.*

A structural alternative

A weaker and more promising idea is that two situations share a meaning not because their futures are identical, but because their futures have the same *shape*, related by some symmetry of the system rather than by outright coincidence.

Definition 9.1 (Structural equivalence). *Fix the metric $d((k, r), (k', r')) := d_{\text{cyc}}(k, k') + |r - r'|$ on $X = \{0, 1, 2, 3\} \times \{0, 1\}$, where $d_{\text{cyc}}(k, k') := \min(|k - k'|, 4 - |k - k'|)$ is the cyclic distance on phase. Let G be a group of isometries of (X, d) acting equivariantly: $g(R_T(x)) = R_T(g(x))$ for every $g \in G$ and every x . States x, y are G -equivalent, $x \sim_G y$, if $y = gx$ for some $g \in G$.*

Remark 9.2. *This is a genuine equivalence relation for the reason a single fixed map is not: reflexivity, symmetry, and transitivity follow automatically from G being a group (the identity, inverses, and closure under composition, respectively), whereas $x \sim_\varphi y \iff y = \varphi(x)$ for one fixed φ need not be symmetric or transitive unless φ itself happens to generate such a group. The equivariance condition is doing real work too: it is what makes $\sim_G$ interact sensibly with Γ at all, since without it G could relate states with unrelated futures for no reason connected to the dynamics.*

Example 9.1 (An isometry that is not a symmetry). *Let $\varphi_1 : (k, r) \mapsto ((k + 1) \bmod 4, r)$, the phase shift. φ_1 is an isometry of (X, d) , since cyclic distance is translation-invariant. Direct computation gives $\varphi_1(R_1(k, r)) = R_1(\varphi_1(k, r))$ for $k = 0$ and $k = 1$, and $\varphi_1(R_1(k, r)) \neq R_1(\varphi_1(k, r))$ for $k = 2$ and $k = 3$: at $k = 0$, $\varphi_1(R_1(0, 0)) = \{(2, 0), (2, 1)\} = R_1(1, 0)$, matching, while at $k = 2$, $\varphi_1(R_1(2, 0)) = \{(0, 0), (0, 1)\} \neq \{(0, 0)\} = R_1(3, 0)$, failing.*

Remark 9.3. *This computation shows φ_1 fails equivariance, and the honest conclusion is narrower than a first pass at this example suggested. Since equivariance must hold everywhere for $\langle \varphi_1 \rangle$ to generate a valid G , and it demonstrably does not (it fails at $k = 2, 3$), φ_1 cannot be used to build a nontrivial structural equivalence on this system at all under Definition 3: the only group of isometries equivariant with this particular Φ across the whole space is the trivial one. What the computation does show, and what is worth keeping, is that φ_1 is a genuine isometry of the state space that happens to agree with the dynamics on part of its domain and disagree on the rest, phases 0, 1 against phases 2, 3, precisely the phases adjacent to clearing. That is a real fact about where the system's transition rule does and does not resemble a uniform relabeling, but it is a fact about a local, not global, correspondence, best formalized by a groupoid of partial symmetries rather than a group if pursued further, an extension not carried out here. A word like “waiting” that treats all four phases alike is, on this reading, not grounded in even a partial group symmetry of the dynamics; it is grounded in nothing more than the coarse observation Π discussed already, and the appeal of the phase-shift computation is diagnostic rather than foundational: it locates precisely where a uniform label stops being defensible, without supplying the label a genuine structural warrant anywhere.*

What a semantic category actually is

Neither $\sim_\Gamma$ nor G -equivalence is what ordinary language means by a category, and this chapter's honest conclusion is that nothing in this book's vocabulary should be expected to be. A semantic category, “red light,” “waiting,” “sleeping,” is generally an observation map Π in the sense of Chapter 6, chosen for reasons external to continuation fidelity entirely: convenience, communicability, whatever distinctions happened to be salient to whoever built the vocabulary. Chapters 2, 3, and 6 did not need this chapter's help to already supply the complete

toolkit for auditing such a choice once made: whether Π is state-sufficient (Chapter 6), and if not, exactly how much it loses and at which fibers (Chapter 3). What this chapter adds is a second, independent question worth asking of the same Π : does it happen to align with a genuine equivariant symmetry of the system, in the sense of G -equivalence, or does it merely resemble one locally, as “waiting” does on the traffic light, agreeing with a candidate isometry on two of four phases while failing the equivariance a genuine group would need everywhere. A category can be lossy and still correspond to a real, if partial, structural pattern; it can also be lossy and correspond to no pattern in the dynamics at all. This book’s machinery can tell these apart even when neither rises to the strength of a genuine group symmetry. It cannot tell you which category to use, because that was never a mathematical question in the first place.

Remark 9.4 (On the original theorem). *The equivalence class construction this chapter was meant to formalize, meaning as continuation-equivalence rather than state-equivalence, survives, but not as a single theorem with one canonical $\cong$. It survives as a choice between at least two precise, inequivalent relations, $\sim_\Gamma$ (too strict for ordinary use) and G -equivalence under an equivariant symmetry group (precise, but demanding enough that this book’s own worked example satisfies it only trivially), together with the recognition that most categories anyone actually uses are neither, and are auditable against both without being reducible to either.*

Chapter 10

Triple Stores and Reachability Closure

A relational table starts from entities and attaches properties to them. A graph database of the kind an RDF triple store implements starts from relations and lets entities emerge as whatever sits at the ends of them. A single fact, Paris is the capital of France, becomes an edge, not a row, and a database built entirely out of edges like this one turns out to behave less like a filing cabinet and more like something this book has spent its length studying: a small set of local rules whose repeated application reaches much further than what was explicitly written down. The temptation is to call this generation in exactly the sense of Chapter 1's fern. That temptation should be resisted, and resisting it properly is most of what this chapter does.

Reachability, not generation

Definition 10.1. For entities X and triples $E \subseteq X \times P \times X$ (with P a set of predicates), the one-step reachability operator is $\Phi(x) := \{y : (x, p, y) \in E \text{ for some } p \in P\}$, and $R_T(x) := \bigcup_{n=0}^T \Phi^n(x)$ is the set of entities reachable within T edges.

Remark 10.1. This is not new. It is the Kauffman correspondence of Chapter 2 of Continuation Geometry [8], already proved there by induction to satisfy $S_{t+1} = S_t \cup \text{Adj}(S_t)$, applied here with $\text{Adj}(S_t) := \bigcup_{x \in S_t} \Phi(x)$. That chapter was careful to distinguish this construction from generation in the sense of a Hutchinson attractor, and the distinction matters enough here to restate: Φ is not, in general, a contraction of anything. There is no metric under which traversing an edge shrinks distances, no completeness argument, no Banach fixed point, and correspondingly no unique limit shape that repeated traversal converges to. $R_T(x)$ simply grows, a monotone union exactly as Proposition 1 of Continuation Geometry's Chapter 2 already proved for this recursion in general. A triple store is an instance of accumulation, not of the contracting machinery Chapter 1 of this book built. Calling it generative in that stronger sense would be the same conflation this book's own first chapter took care to rule out.

Remark 10.2. What survives of the analogy is real, and worth keeping separate from what does not. A small, explicit generator set producing a much larger implicit structure is common to both an IFS attractor and a triple store's transitive closure; only the mechanism differs, metric contraction to a fixed point in one case, combinatorial branching outward from a seed in the other. The next section makes the branching case precise on its own terms rather than borrowing Chapter 1's vocabulary for it.

This book	RDF / graph databases
one-step reachability Φ	edge traversal
$R_T(x)$	T -hop neighborhood
continuation volume Vol_T	neighborhood size
$\sim_\Gamma$	indistinguishable neighborhoods
continuation quotient	graph compression
sufficient statistic (Chapter 5 of <i>Continuation Geometry</i>)	context coordinate
continuation distance d_Γ	embedding similarity

Remark 10.3. *This table is a reading aid, not a further claim; every row is established, corrected, or left explicitly open in the sections that follow, and none of them should be taken as settled by appearing in a table.*

How much is implied by how little

Proposition 10.1 (Reachable volume under bounded branching). *If $|\Phi(x)| \leq b$ for every x , then $|R_T(x)| \leq \sum_{k=0}^T b^k$.*

Proof. By induction on T . $|R_0(x)| = 1 \leq b^0$. Assume $|R_T(x)| \leq \sum_{k=0}^T b^k$. Every element of $R_{T+1}(x)$ is either already in $R_T(x)$ or reached by one further edge from some element of $\Phi^T(x)$, so $|R_{T+1}(x)| \leq |R_T(x)| + |\Phi^T(x)| \cdot b \leq |R_T(x)| + b^{T+1}$ (bounding $|\Phi^T(x)|$ by b^T , itself an immediate consequence of the same bound applied one level down), giving $|R_{T+1}(x)| \leq \sum_{k=0}^{T+1} b^k$. $\square$

Corollary 10.1. *For n entities each with out-degree at most b , edge storage is $O(nb)$, while a single entity's T -step reachable volume can be as large as $\Theta(b^T)$, exceeding total edge storage once $T > \log_b(n)$ roughly.*

Remark 10.4. *This is the honest version of the compression claim: not that the store is somehow secretly larger than it appears, but that the quantity worth measuring, T -step reachable volume in the sense of Chapter 6's Vol_T , specialized here to counting measure on a discrete graph, grows combinatorially in T while storage grows linearly in n . A social graph storing a hundred million edges can have individual T -step neighborhoods, for modest T , larger than the edge set itself. Nothing about this requires the store to compute or cache those neighborhoods in advance; Φ alone, applied repeatedly at query time, is enough.*

Context coordinates as retained history

A triple store proper cannot represent two facts about the same subject and predicate at once. A quad store, storing (s, p, o, g) for a context or graph label g , can. Whether dropping g is a representational loss or an operational one, in the sense of Chapter 10, depends on which kind of store is being asked.

Example 10.1 (Representational: a query drops context). *A quad store holds both $(\text{Alice}, \text{worksAt}, \text{Acme}, 2024)$ and $(\text{Alice}, \text{worksAt}, \text{BetaCorp}, 2026)$. A query that ignores g sees both employers as simultaneously true, losing the temporal distinction, but the underlying store still holds both facts and a differently written query recovers them. This is Chapter 3's fiber loss, not Chapter 11's operational loss: nothing was destroyed, only misread by one particular query.*

Example 10.2 (Operational: the schema has no g slot). *A plain triple store with a functional predicate, at most one object per subject-predicate pair, cannot hold both facts at all; inserting the second triple overwrites or conflicts with the first. Here the distinction is gone in the sense Chapter 11 gives that word: no query against this store, however cleverly written, recovers the discarded employer, because the schema itself never retained the information needed to answer it.*

Corollary 10.2 (Context coordinates are sufficient-statistic augmentation). *Let X be the space of (s, p, o) triples and let g range over contexts. If $\Gamma(s, p, o, g) \neq \Gamma(s, p, o, g')$ for some s, p, o and $g \neq g'$, then the plain-triple observation $\Pi(s, p, o, g) := (s, p, o)$ is state-insufficient in the sense of Chapter 6, and the quad-store state (s, p, o, g) is exactly the augmentation $x \mapsto (x, \varphi(H_t))$ of Chapter 5 of Continuation Geometry, with g playing the role of the retained coordinate $\varphi(H_t)$.*

Proof. This is Chapter 5 of Continuation Geometry instantiated directly: that construction augments a state x with a summary $\varphi(H_t)$ of history exactly when Γ depends on more than x alone, and requires nothing beyond checking that dependence, which is the stated hypothesis $\Gamma(s, p, o, g) \neq \Gamma(s, p, o, g')$ here. No new argument is needed; the triple store’s context label is an instance of an already-proved construction, not a new one. $\square$

Remark 10.5. *The graph label g , read this way, is a sufficient statistic in the sense of Chapter 5 of Continuation Geometry: it is exactly the augmented coordinate that construction called for, x silently becoming (x, g) , when a single observable state is not enough to determine what comes next. A triple store without context coordinates is a system that has already discarded that augmentation; whether the discarding is fixable is not a fact about the data remaining somewhere unread, it is a fact about whether the schema was built to keep it.*

Meaning, corrected

Remark 10.6. *It is tempting, watching graph embeddings work well in practice, to write $\text{Meaning}(x) \approx \Gamma(x)$ and stop there. Chapter 9 already showed why the naive version of that claim, meaning as literal reachable-set equality, is too strict: two entities with almost, but not exactly, identical neighborhoods would count as having nothing in common, which is not how embeddings or ordinary semantic judgment actually behave. What graph embeddings compute is closer to a distance than an equality, neighborhoods aggregated and compared by similarity rather than checked for exact coincidence, which is Chapter 6’s d_Γ , not $\sim_\Gamma$. Two entities being semantically close because their T -step neighborhoods are near in Hausdorff distance is a defensible, gradeable claim in a way that Chapter 9 already proved bare equality is not. The correction Chapter 9 made for a four-state traffic light applies without modification to a hundred-million-edge graph.*

Remark 10.7 (The shape of the connection to graph neural networks, and its limit). *The structural form $\text{Embedding}(x) := F(R_T(x))$, for some aggregation F , names the right correspondence: an embedding is a compressed coordinate on a continuation space, exactly matching this book’s d_Γ once F is chosen so that similar aggregated outputs correspond to nearby reachable sets. What this remark does not claim is that graph neural networks compute this F mechanically as stated. Real architectures aggregate learned feature vectors attached to nodes through differentiable, trained functions, layer by layer, not a fixed F applied once to the raw reachable set $R_T(x)$; the correspondence here is a claim about shape, an aggregation of neighborhood structure into a compressed coordinate, not a claim that this book’s formalism reproduces the mechanics of message passing or attention. Treating $F(R_T(x))$ as*

a structural analogy for what an embedding is doing, rather than a specification of how one is computed, is the honest version of this connection.

What compresses, made exact

Corollary 10.3 (Graph quotient). *Fix horizon T . The relation $x \sim_T y \iff R_T(x) = R_T(y)$ is an equivalence relation on graph nodes, and the quotient graph it induces is the coarsest graph preserving every T -bounded reachability query against the original.*

Proof. $\sim_T$ is $\sim_\Gamma$ specialized to $\Gamma := R_T$, so it is an equivalence relation by the definition of Chapter 2. That the induced quotient is coarsest among those preserving T -bounded queries is Theorem 1 of Chapter 3 (the coarsest zero-loss quotient), instantiated with H the node set and $\Gamma := R_T$: any coarser identification merges some pair with $R_T(x) \neq R_T(y)$, which some T -bounded query distinguishes, by definition of R_T as the record of every node reachable within T edges. $\square$

Remark 10.8. *Graph compression by neighborhood equivalence is not analogous to a continuation quotient. It is one, with no translation required beyond choosing $\Gamma := R_T$.*

Remark 10.9 (An unplanned validation of Chapter 6). *Graph practitioners already draw, informally and by necessity, the same distinction Chapter 6 built formally: a node's identifier and a node's neighborhood are different things, and two nodes can share one while differing in the other. A recommendation system that deduplicates users by neighborhood similarity, ignoring stated identity, and a knowledge graph that keeps two identically-labeled entities apart because their surrounding facts differ, are both already committed to the claim Chapter 6 had to argue for from first principles: that an observable label need not be sufficient to determine what a node's structure actually is. This does not make Chapter 6's argument redundant. It makes it independently confirmed, by a field that arrived at the same distinction for entirely practical reasons and never needed to call it state insufficiency to rely on it.*

Closing

A triple store is a real system built, apparently independently of this book's concerns, around the same primitive this book has spent its length formalizing: not the state of an entity, but what remains reachable from it. Where the analogy to Chapter 1's attractors was tempting and wrong, this chapter said so and pointed to the correspondence that is actually apt, Chapter 2's accumulation. Where the analogy to Chapter 9's meaning was tempting and too strong, this chapter softened it to the distance Chapter 6 already built. What needed no correction at all, the compression argument and the context-coordinate reading, is offered as this chapter's genuine contribution: not a new theorem, but two places where an existing one turned out to already be about something built for entirely different reasons a decade before this book was written.

Chapter 11

Representational Loss versus Operational Loss

Tell someone that a system “cleared its memory” and they will picture one of two entirely different events, and ordinary language gives them no way to know which. In the first, a record-keeper simply starts describing two situations the same way going forward, a paperwork decision that changes nothing about the world the paperwork describes. In the second, something is physically erased, and the capability that depended on it is gone regardless of what anyone writes down afterward. A summary of this book’s own traffic-light example blurred exactly this line, narrating a change of label as though it were a change of wiring, and the blur is worth resolving properly, since the two events have the same size in bits and opposite consequences in the world.

Two kinds of forced merge

Definition 11.1 (Representational loss). *For a quotient $q : H \rightarrow O$, the representational loss at a fiber o is $\mathcal{L}(o) := \log |C_\Gamma(o)|$, exactly as in Chapter 3: the log-count of distinct continuation spaces a description choice groups under one label, with the underlying dynamics Φ left untouched.*

Definition 11.2 (Operational loss). *For an intervention T (Chapter 7) applied on a set S , the operational loss on S is $\mathcal{L}_{\text{op}}(T, S) := \log |\{\Gamma(x) : x \in S\}|$, computed on the pre-intervention values of Γ : the log-count of distinct futures S ’s members actually had before T forced them to a common target.*

Remark 11.1. *The two definitions look identical, and the resemblance is the point, not an accident to apologize for. What differs is not the formula but what it is being asked about. Representational loss is a property of a description laid over an unchanged system: the pedestrian’s request is still there, still shaping the real controller’s real next state, and only the label an observer uses to talk about it has become too coarse to reveal that. Operational loss is a property of the system after something has actually acted on it: the request itself is gone, and no amount of relabeling recovers it, because there is nothing left to relabel accurately.*

What Chapters 2 and 3 actually did

Proposition 11.1. *Every computation in Chapters 2 and 3 involving the traffic light’s alternative clearing rule used a single fixed Φ throughout; $\mathcal{L}_{\text{op}} = 0$ everywhere in that material, since no intervention was ever applied.*

Proof. The alternative rule considered in Chapter 2 (clearing whenever currently at phase 0, rather than on arrival) was introduced and evaluated entirely as a candidate quotient q , compared against the one fixed transition function Φ given in that chapter’s definition. Chapter 3’s loss computation, likewise, computed $C_{\Gamma}(o_{\text{bad}})$ using R_1 values generated by that same single Φ . At no point was a second transition function constructed, nor was any operator applied to a state to force its subsequent behavior; every quantity computed was a property of a label held up against one unchanging system. $\square$

Remark 11.2. *This is worth stating as a proposition rather than a footnote, because the natural way to narrate the same material out loud tends to drift toward the operational reading without anyone intending it to. Saying a design “forcibly changes state A into state B ” is a perfectly natural way to describe a coarse label, but it is also, read literally, a description of an intervention, and the two readings support different conclusions. Under the representational reading, nothing about the intersection’s actual behavior was ever at stake; the pedestrian’s request was tracked correctly by Φ the entire time, and the only failure was in how a hypothetical observer using the bad label would have talked about it. Proposition 11.1 is the precise statement that this is, in fact, all that was shown.*

The operational twin

Example 11.1 (The same bit, actually erased). *Let T' be the operator sending $(0, 1) \mapsto (0, 0)$ (and fixing every other state), a genuine intervention in the sense of Chapter 7: it forces the physical request bit to 0 at the moment of intervention, not merely relabels the situation.*

Proposition 11.2. $\mathcal{L}_{\text{op}}(T', \{(0, 0), (0, 1)\}) = \log 2$, exactly the value Chapter 3 computed for the representational case.

Proof. The pre-intervention values are $\Gamma(0, 0) = R_1(0, 0) = \{(1, 0), (1, 1)\}$ and $\Gamma(0, 1) = R_1(0, 1) = \{(1, 1)\}$, already established as distinct in Chapter 6. So $|\{\Gamma(0, 0), \Gamma(0, 1)\}| = 2$, giving $\mathcal{L}_{\text{op}} = \log 2$. $\square$

Remark 11.3. *The number is identical to Chapter 3’s one bit, and this identity of magnitude is exactly what makes the two cases easy to conflate and important to keep separate. Under T' , a pedestrian who ran up and pressed the button while the light sat at phase 0 genuinely loses the system’s acknowledgment of that press: the controller’s actual future behavior is now the same whether or not the button was pressed, because the operator overwrote the state that would have carried the difference forward. This is not a labeling failure a better catalog could fix. The information a better label would have preserved is no longer there to preserve.*

Why the difference is not academic

Remark 11.4. *A purely representational loss is repaired for the cost of a better label, since Proposition 11.1’s kind of failure never touched the underlying system; recomputing $\sim_{\Gamma}$ correctly and using*

it instead costs nothing beyond noticing the old label was wrong. An operational loss, once incurred, is repaired only by whatever it takes to restore the erased distinction physically, if that is even possible, since Chapter 7's machinery already established that an intervention forcing $\Gamma(T(x))$ to a fixed target is not, in general, invertible by any further relabeling. Mistaking the second kind of failure for the first is not a rounding error. It is the difference between believing a problem is free to fix and not noticing that it no longer can be.

Chapter 12

Closing

The word in this book's title was chosen before most of the book was written, which is the ordinary way titles get chosen and also, on reflection, a reasonable place to end by checking. A fern grown from four contractions is generative in the most literal sense available: nothing was accumulated, everything was produced by rules acting on their own output. Not everything that came after Chapter 1 earned the word as cleanly, and an honest closing chapter has to say so rather than let the title do the arguing for chapters that moved on to something else.

What was established

Stated plainly: this book proved the Continuation Attractor Theorem in full, a genuine Hutchinson-Banach result applied to the Hausdorff-metric space of reachable sets already built in the first book, with a worked, verified Cantor-set instance. It defined the continuation quotient without circularity, having first caught and discarded an earlier attempt that assumed its own conclusion, and proved its universal property and its invariance property from that definition alone. It proved that the coarsest quotient losing no continuation-relevant information is $\sim_\Gamma$ itself, tying directly back to the sufficient-statistic material of the first book's Chapter 5 rather than introducing a competing notion. It computed, exactly and by verified arithmetic rather than by hand-wave, where a real toy system's design was provably correct and where a superficially similar alternative would have provably destroyed information, and it caught a genuine numerical error in its own first attempt at that computation before publishing the corrected version. It placed generation and collapse in one vocabulary, a continuation operator, and proved that vocabulary closed under composition regardless of where an operator sits on the genuinely separate axes of contraction and injectivity, catching and correcting an overclaim of its own that had compressed those axes into one number. It proved a real Collage Theorem, three lines long and built entirely from material the first chapter had already established, and used it to prove that every compact target is approximable, while keeping that result honestly separated from the much harder conjecture it is often mistaken for. It resolved an explicitly flagged gap, state insufficiency, without inventing a new toy system to do it, by observing an already-built one differently. It gave staleness and invariant-block separation different proofs and different remedies rather than one gesture toward both, reusing a Grönwall estimate from the first book's own appendix rather than deriving a new one. It showed that investment-movable admissibility needed no new operator definition at all, only

a decision to point Chapter 4's machinery at constraint sets instead of shapes. It resolved its own explicitly flagged undefined symbol, settling on two precise and mutually irreducible notions of continuation-equivalence rather than one overloaded one, and said plainly when the answer reduced to material already on hand. It found, in an industrial graph database built for entirely unrelated reasons, a genuine instance of this book's own primitive, and was honest that the instance fit one chapter's vocabulary well and another's not at all, correcting a tempting but wrong analogy to generation before it could take root. And, close to the end, it caught a conflation in a plausible-sounding retelling of its own second chapter and gave the conflation a name and a formal treatment rather than a footnote.

What was not

The Universal Continuation Generation conjecture, named precisely in Chapter 5, was not proved, and this book's own methods do not reach it; the gap between existence of an approximating family and existence of a single universal one was made exact, not closed. Whether a genuinely dimensional, rather than merely cardinality-based, obstruction to finite sufficient statistics exists remains exactly where the first book left it, untouched here. The question of whether a bounded, monotone, expansive operator family converges to a well-defined limit from below, Chapter 8's open problem, was demonstrated once and proved never. Whether any operator actually named Collapse, Refuse, or Project elsewhere in this corpus is literally, technically a contraction or a quotient in the senses built here was repeatedly flagged as an empirical question about those specific constructions and was never once investigated. The nested-constraint account of agency raised during this book's planning was set aside for lack of any formal target this vocabulary could give it, and it should stay set aside until one is found rather than be forced into whichever chapter seemed most convenient. And no chapter here built a general repair theory for operational loss analogous to the first book's account of repair as capture; Chapter 10 named the distinction between a loss that can be undone by relabeling and a loss that cannot, but said nothing about how, or whether, the second kind is ever actually recovered.

Was generative the right word

Read the table of contents cold and the honest pattern is this: two chapters, the first and part of the eighth, are generative in the sense the title promises, operators building structure up from nothing but their own repeated application. A comparable number, the second, third, sixth, seventh, ninth, tenth, and eleventh, are about the opposite motion: quotients collapsing distinctions, losses computed and located, insufficiencies made explicit, failure modes kept formally apart, conflations caught and corrected, an industrial example's own generative-sounding analogy corrected on arrival. The fourth chapter is the hinge, though its own closing claim had to be corrected before it could bear the weight this paragraph puts on it: generation and collapse are not two regions of one Lipschitz parameter, a single number never separated them, only injectivity did that, independently of contraction. What the fourth chapter actually earned is narrower and still real: both motions share one vocabulary, continuation operators, and are checkable by the same closure property regardless of which region of that richer, multi-axis space they occupy. The book was never really about generation any more

than it was about collapse. It was about the vocabulary both turned out to need.

A fourth thread the count above simplifies

The two-bucket count above, generative against everything else, is honest as far as it goes, but it is not the finest division available, and pretending otherwise would repeat exactly the kind of premature compression this book has spent its length arguing against. Chapters 7 and 8 do not answer the question the rest of the book keeps returning to: whether two histories, states, or descriptions can be treated as the same without losing anything continuation-relevant. Invariant decomposition is not a quotient in that sense at all; it says nothing about identifying x with y , only that some region of the state space is causally sealed off from the rest, a fact about a set, not a relation on a pair. Staleness and intervention are not equivalence questions either; they concern what happens to a fixed piece of knowledge, or a fixed state, as time elapses or as force is applied, a question about change over time rather than about which things coincide. Chapter 8's investment operators inherit the same character: a boundary moving as effort accumulates is a claim about a schedule, not about which states are interchangeable.

What Chapters 7 and 8 share, and what generation, quotienting, and sufficiency alike do not quite have room for, is a concern with time and force themselves: how a fact about a system degrades, is confined, or is overridden as the clock runs or as effort is spent, independent of whether two things were ever the same to begin with. This book is better described by four threads than by two: generation, quotienting, sufficiency, and the dynamics of admissibility and knowledge under time and intervention. Naming the fourth one now, rather than folding it into whichever of the first three it resembled most, is the same discipline this book applied to its own Lipschitz overclaim in Chapter 4 and its own undefined equivalence in Chapter 9: a count that comes out even is not evidence the count was right.

Remark 12.1. *If this book has a real center of gravity, it is not the fern and it is not the confluence of rivers. It is the discovery, made honestly and only in retrospect, that a project which set out to explain how futures get built spent most of its actual length explaining how they get lost, mislabeled, confined, staled, and mistaken for one another, and that neither half makes sense as a theory on its own. A theory of generation with nothing to say about loss would not have been able to tell a correct design from an incorrect one in Chapter 2. A theory of loss with nothing to say about generation would have had no attractor to measure the loss against in the first place. The title stays, because the object it names, a continuation operator, really does generate both halves from one definition. But anyone reading this book for a theory of how futures are built should know, going in, that they are about to spend at least as much time learning how futures are taken away.*

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